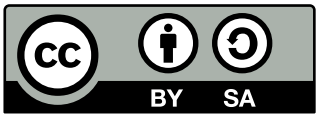
Mathematical Programming with Julia

An open-source approach to
Linear & Mixed Integer Programming
Version 1.3
Julia 1.8.3 (at least)/JuMP 1.0 (at least)

Stefan Røpke, Evelien Van Der Hurk, Richard Lusby & Thomas Stidsen

VERSION 1
2026

ISBN: 978-87-93458-25-3
Publisher: DTU Management, Technical University of Denmark, Denmark, 2022



The authors, Richard Lusby & Thomas Stidsen, would like to acknowledge

Thomas would like to thank Charlotte

Richard would like to thank Angelika

1 Welcome to the course: Course plan

Welcome to the course. This book/pdf-file contains a brief installation chapter, a brief introduction chapter on Linear Programming, a brief chapter on Mixed Integer Programming, and all the exercises of the course. The main body of this book is the Linear programming assignments and Mixed Integer Programming assignments.

In this course you will learn to formulate mathematical optimization models, and write computer programs in Julia to solve them. Lectures will be provided on-site only. Also, support for exercises takes place on-site. Lectures take place in Aud. ??? in building ???. Workshops take place in Building ???, rooms: ??? (main room, with blackboard list for TA's), ???, ??? and ???, where TAs are available for help you with the exercises.

In this course you will learn to create models and implement them into computer programs in Julia, so that you can find their optimal solution. Notice: We will not stream the lectures. If you want to hear any of the lectures, you have to come to Aud. ??? in building ??? at the right time (see below). We will make the slides available the day before the lecture. Every day also consists of a workshop part where you work independently on exercises. TA's are available for help in our main exercise room ??? Building ???. You can also always request an exercise or problem to be reviewed in the next lecture. There will be no help on-line, only on-site.
As preparation before the course starts install Julia/JuMP and an editor to write programs, e.g. VS Code.

- For each day of the course, except the last examination day and the catch up day (22/1 2026).
- In the end of each day, we will upload the correct answers to all assignments of the day, 17.00 to the content for that day in DTU-learn: `https://learn.inside.dtu.dk/d2l/home/241623`.

This year we are experimenting with some more lecture time. However, the focus of the course is learning by *doing*. Therefore these lectures will take the form of "guided workshops", which means that short explanations will be followed by hands-on exercises that are immediately followed up by feedback on the results. Throughout the course,

you will gradually work more independently and we will reduce the amount of lecture time.

Notice: If you feel you need to brush up on Linear Programming, we refer to Chapter 3. In Chapter 7 Mixed Integer Programming is briefly reviewed.

1.1 Week 1

Monday 5/1

1. Lecture 1: Course introduction and Linear Programming (Thomas)
2. Jewellery Production 4.1
3. Micro Brewery 4.2
4. Blending 4.3

Tuesday 6/1

1. Lecture 2: Basic Julia/JuMP (Stefan)
2. Blending 2 4.4
3. Factory Planning 4.5
4. Project Schduling 4.6

Wednesday 7/1

1. Lecture 3: Mixed Integer Programming (Thomas)
2. Young Couple 6.1
3. Startup Fund 6.2
4. Logic 6.3
5. Stamps 6.4, data available on DTU-learn.

Thursday 8/1

1. Lecture 4: MIP hardness (Stefan)
2. Micro Brewery 2 6.5
3. Santa's Workshop Tour 2019 6.6, data available on DTU-learn.
4. Factory Planning 2 6.7

Friday 9/1

1. Lecture 5: AI for modelling (Thomas)
2. Student Teacher Meetings 6.8, data available on DTU-learn.
3. Workplan for Teaching Assistants 6.9, data available on DTU-learn.
4. Scrap Removal 6.10

1.2 Week 2

Monday 12/1

1. Lecture 6: Network models (Stefan)
2. Shortest Path [new]
3. Min cost flow [new]
4. Multicommodity flow [new]
5. Network design [new]

Tuesday 13/1

1. Lecture 7: Vehicle routing (Stefan)
2. Groceries 6.15
3. Tennis 6.14
4. Aircraft Landing 6.16

Wednesday 14/1

1. Lecture 8: Energy models (Mikkel)
2. Tariff Rates 6.12
3. Railway Depot Shunting 6.13, data available on DTU-Learn

Thursday 15/1

1. Food Festival 6.17, data available on DTU-learn.
2. The Wedding Planner 6.18, data available on DTU-learn.

Friday 16/1

1. Hot Air 6.21
2. Wind Farm Cable Routing 6.22, data available on DTU-Learn

1.3 Week 3

Monday 19/1

1. Lecture 9: Math heuristics (Mikkel)
2. Wedding Planner, math-heuristic algorithms 6.19
3. Stamps, math-heuristic algorithms 6.20

Tuesday 20/1

1. Lecture 10: Multi-Objective Mixed Integer Programming (Thomas)
2. Multi-objective Startup fund 8.1
3. Multi-objective Young Couple 8.2
4. Multi-objective Wedding planning 8.3
5. Multi-objective Aircraft Landing 8.5

Wednesday 21/1

1. Lecture 11: The exam (Thomas)
2. Exam test (old exam)
3. University Timetabling 6.23, data available on DTU-learn.

Thursday 22/1

1. Catch up day

Friday 23/1

Exam day

Contents

[1] Taken from Williams: "Model Building in Mathematical Programming"
[2] Taken from Williams: "Model Building in Mathematical Programming"

[3] Taken from Williams: "Model Building in Mathematical Programming"
[4] Hillier and Lieberman: Introduction to Operations Research"
[5] This comes from the traditional Christmas optimization challenge, see: https://www.kaggle.com/c/santa-workshop-tour-2019
[6] Taken from Williams: "Model Building in Mathematical Programming"
[7] Taken from Williams: "Model Building in Mathematical Programming"

2 Installation

This book focuses on Mathematical Programming modelling, and we will use Julia and JuMP as the vehicle to implement the models. This is necessary, since these models cannot be solved analytically. We must emphasize that this book is **not** about Julia programming. For this, there are a number of other resources available at `www.julialang.org`. JuMP is a software package which makes it easy to implement Mathematical Programming models in Julia. In this chapter, we will briefly describe how to install Julia, JuMP, selected solvers, and software editors.

2.1 Julia

Julia is a new programming language, initiated in 2009, which is "as fast as C and as easy as Python". Installing Julia is easy, simply download the correct version for your computer from `https://julialang.org/downloads/` and follow the instructions. In this course we will use version v1.8.3 of Julia, which is the newest version. If you have a slightly older version, e.g. v1.7, you do not have to reinstall Julia. Julia supports Windows, Mac, Linux, and FreeBSD.

2.2 JuMP

When Julia is installed, installing JuMP is easy: Start Julia in a command-line. In the command-line write: import Pkg; Pkg.add("JuMP"). Then the JuMP package will be automatically installed. This may take a few minutes.

2.3 Solvers

Given a Mathematical Programming model, either a Linear Programming (LP) or Mixed Integer Programming (MIP) solver is necessary. In this course our standard solver is HiGHS, which is open-source. If you already have Gurobi installed this can also be used.

The JuMP modelling package however supports a number of solvers, both commercial and open-source:

1. HiGHS: **open-source** solver, for both LP models and MIP models
2. Gurobi: Commercial industry leading solver, for both LP models and MIP models
3. Cplex: Commercial industry leading solver, for both LP models and MIP models
4. Clp: **open-source** solver for solving LP models. Not further developed
5. GLPK: **open-source** solver for solving both LP models and MIP models. Not further developed
6. Cbc: **open-source** solver for solving MIP models. Not further developed
7. Xpress: Commercial industry leading solver, for both LP models and MIP models

In this book, we recommend to use either HiGHS, Gurobi, or CPLEX. HiGHS is an opensource solver which is just as easy to install as JuMP and can be used for free. All the models in this book can be solved with the best open-source solver, HiGHS, if the model is implemented correctly. The two commercial solvers Gurobi and CPLEX are state-of-art solvers and can be obtained for academic usage for free, but they are harder to install and can then only be used for teaching and research. But they are significantly faster for most MIP models.

There is a longer list of supported solvers shown in the JuMP documentation `https://jump.dev/JuMP.jl/stable/installation/` and the list keeps growing. We have picked what we consider the best solvers for use in this book above. Furthermore, a number of the supported solvers are for other types of mathematical models not considered in this book: QP (Quadradic Programming), NLP (Non-Linear Programming), (MI)NLP (Mixed Integer Non-Linear Programming), SOCP (Second-Order Cone Programming), SDP (Semidefinite programming), and several others.

2.4 Editors

The Julia/JuMP models are in principle simple ASCII text files, and any editor can be used to manipulate them, even Windows Notepad can be used. The models can then be executed from the command-line. We recommend the use of Visual Studio Code, `https://code.visualstudio.com`.
Installing these environments, both of which are open-source, and connecting them to Julia can sometimes be tricky, depending on your computer setup.

2.5 Why Julia ?

The material in this book is part of the material used in our course "Mathematical Programming Modelling" (42112), taught at the Technical University of Denmark. This course about making LP models and MIP models has been taught at our university for more than 40 years, the last 18 years by us. For the entire period until 2018 the specialized Mathematical programming language GAMS `www.gams.com` was used. Since 2019 we have used Julia/JuMP. Using Julia/JuMP comes with a number of advantages:

- The whole system is open-source, when using the HiGHS open-source solver, i.e. free of charge to use.
- Integration with other programs is simpler, because these can simply be programmed in Julia.
- Like GAMS, multiple different solvers can be applied, and easily be switched between different solvers.

Our experience with Julia/JuMP since 2019 has been so good that we consider the days of dedicated Mathematical Programming languages like GAMS, AIMMS, AMPL, MPL to be over. Julia/JuMP is now the only programming language used in the courses at the Operations Research section, at the Department of Management Engineering. It is being used in all our 10+ courses on Operations Research for, e.g., meta-heuristics, Dantzig-Wolfe decomposition, Benders Decomposition and Network optimization.

3 Linear Programming

LP is a classic technique in Operations Research and was invented independently several times during World War II. With the invention of the Simplex algorithm, by George Dantzig Danzig (1947), the models could also be optimized, i.e. the best solution values of the decision variables given an objective and a set of constraints. The optimization algorithms for LP is not the topic of this book, in Chapter 10 we will give links to literature on this important topic. Here we focus on **modelling** problems with the LP model. We will now introduce LP models using the two first assignments from Chapter 4, "Incredible Chairs" and "Incredible Chairs II".

3.1 Incredible Chairs

The company Incredible Chairs produces two different types of chairs, A and B, where one unit (pallet of chairs) of Chair A can be sold for a profit of 4 and one unit of Chair B can be sold for a profit of 6. The chairs are produced on three production lines:

1. Line 1 requires 2 hours to produce one unit of Chair A, and at most 14 hours can be used due to other production on the line.

2. Line 2 requires 3 hours to produce one unit of Chair B, and at most 15 hours can be used.

3. Line 3 requires 4 hours to produce one unit of Chair A and 3 hours to produce one unit of chair B, and at most 36 hours can be used.

Suppose you are the production planner at the company, how will you plan the production of the chairs? The first important thing to identify is what you are in control of. In other words, what are the decisions you can make as production planner? In this case, you can decide **how many** of each type of chair to produce. Since these quantities are unknown and must be decided, we define two **decision variables**, $x_A \geq 0$ and $x_B \geq 0$, which will store the quantity of each type of chair. Notice that the variables have a domain, i.e. an interval of feasible values. Here, they are each constrained to take a positive value. The job for you as the production planner is to find the best possible

values for these two variables.

There are many possible solutions represented by the decision variables (x_A and x_B). As the production planner, you are, however, not interested in a random production plan, stated in the variables x_A and x_B, but you want the **best feasible** production plan. To compare the different solutions, an **objective function** $f(x_A, x_B)$ which quantifies the utility of the solution is used. For this example, the objective function measures the profit of the solution obtained by producing x_A and x_B chairs. The objective function can therefore be stated as:

$$\max f(x_A, x_B) = 4 \cdot x_A + 6 \cdot x_B$$

Not all solutions are possible; they have to be **feasible**. This means that the values assigned to the variables should satisfy the restrictions that are imposed by the production lines. In particular, for each production line, the planned production of chairs cannot exceed the resource limit available. On line 3, the sum of the required resources cannot exceed 36. Given the unknown values of the decision variables, we can now formulate an equation - a **constraint** - that limits the values of x_A and x_B to values values which are possible given the limitations on line 3:

$$4 \cdot x_A + 3 \cdot x_B \leq 36$$

Similarly, there are two other constraints imposed by lines 1&2. They are given below:

$$\begin{aligned} 2 \cdot x_A &\leq 14 \\ 3 \cdot x_B &\leq 15 \end{aligned}$$

The complete LP model is therefore:

$$\begin{array}{lrcrcr} \texttt{Maximize.} & 4 \cdot x_A & + & 6 \cdot x_B & & \\ \texttt{Subject to :} & & & & & \\ & 2 \cdot x_A & & & \leq & 14 \\ & & & 3 \cdot x_B & \leq & 15 \\ & 4 \cdot x_A & + & 3 \cdot x_B & \leq & 36 \\ & x_A & & & \geq & 0 \\ & & & x_B & \geq & 0 \end{array}$$

LP models are **linear**, i.e., the only allowed calculations in objective function and constraints are multiplications between constants and decision variables, and the addition and subtraction of variables.

One of the reasons for the popularity of LP models is that today there are a number of efficient mathematical optimization algorithms - *solvers* - which can be used to **solve** (i.e., find the optimal solution to) the LP model. Solvers optimize the mathematical model and in doing so find the best possible feasible values for the decision variables (here xA and xB). To solve the model, it must be loaded into a solver. In this book, we use the standard programming language Julia in combination with the specialized mathematical programming modelling package JuMP. The syntax is different to the above mathematical model but very similar. Below we list the compact version, in Julia/JuMP.

Compact IC model version:

```
1 using JuMP
2 using HiGHS
3 IC = Model(HiGHS.Optimizer)
4
5 @variable(IC,xA>=0)
6 @variable(IC,xB>=0)
7 @objective(IC, Max, 4*xA+6*xB)
8 @constraint(IC, 2*xA <= 14)
9 @constraint(IC,3*xB <= 15)
10 @constraint(IC,4*xA+3*xB <= 36)
11 print(IC)
12 optimize!(IC)
13 println("Termination status: $(termination_status(IC))")
14 if termination_status(IC) == MOI.OPTIMAL
15     println("Optimal objective value: $(objective_value(IC))")
16     println("xA: ",value(xA))
17     println("xB: ",value(xB))
18 else
19     println("No optimal solution available")
20 end
```

It should be clear that despite the different syntax there is a one-to-one correspondence between the LP model given above and the Julia/JuMP model on lines 5 to 10. Notice that the LP model assumes that chairs can be produced in any number, including fractional amounts. In fact, the optimal production plan is to produce 5.25 of chair A and 5 of chair B. If we impose the requirement that $x_A \in Z^+$, i.e. that only **integer** amounts of chairs can be produced, the model becomes a MIP model. This is described further

in Chapter 7.

When we execute the model in Julia, we get the following printout:

```
Max 4 xA + 6 xB
Subject to
 2 xA <= 14.0
 3 xB <= 15.0
 4 xA + 3 xB <= 36.0
 xA >= 0.0
 xB >= 0.0
Presolving model
1 rows, 2 cols, 2 nonzeros
1 rows, 2 cols, 2 nonzeros
Presolve : Reductions: rows 1(-2); columns 2(-0); elements 2(-2)
Solving the presolved LP
Using EKK dual simplex solver - serial
  Iteration        Objective     Infeasibilities num(sum)
          0     0.0000000000e+00 Ph1: 0(0) 0s
          1    -5.1000000000e+01 Pr: 0(0) 0s
Solving the original LP from the solution after postsolve
Model   status      : Optimal
Simplex   iterations: 1
Objective value     :  5.1000000000e+01
HiGHS run time      :          0.00
Termination status: OPTIMAL
Optimal objective value: 51.0
xA: 5.25
xB: 5.0
```

The model is output in first seven lines as per the instruction line 11 of the Julia/JuMP program. This can be beneficial when debugging a model, but is only ever relevant for relatively small models. The following fourteen lines of output comes from the HiGHS solver. The program then prints out that the optimal solution has been found (line 13 in the program) and then finishes by printing the optimal, maximal profit (51.0) (line 15), and the optimal values of the two decision variables, $xA = 5.25$ and $xB = 5.0$ (line 16 and line 17).

3.2 Incredible Chairs 2

What if the company were not producing two different chairs on three production lines, but say ten different chairs on five different production lines? If the same approach were used, the size of the LP model would increase, and it would be quite tedious to input using the direct approach above. In this case, we can formulate a more abstract model. Given a set of chairs C, indexed by c, and a set of production lines P, indexed by p, we can define a (one-dimensional) vector $Profit_c$, which states the profit for each chair $c \in C$, a (one-dimensional) production capacity vector $Capacity_p$, which states the production capacity of each production line $p \in P$, and a (two-dimensional) matrix $RecourseUsage_{p,c}$ that states how much of the available resource on production line p is needed for production of one unite of chair c.

This gives the following LP model:

$$\begin{array}{lrll} \texttt{Maximize.} & \sum_{c \in C} Profit_c \cdot x_c & & \\ \texttt{Subject to :} & & & \\ & \sum_{c \in C} RecourseUsage_{p,c} \cdot x_c & \leq & Capacity_p \quad \forall p \in P \\ & x_c & \geq & 0 \quad \forall c \in C \end{array}$$

The above model is a very **generic** (chair) production model for profit maximization for any number of chairs and any number of production lines. In fact, any (simple) profit maximizing production problem with a number of products and production resources can be modelled in this way. In the remainder of this book, we will consider two types of variables: **direct variables** and **abstract variables**. The above model has 10 $(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10})$ direct variables, but only 1 abstract variable (x_c). Correspondingly, we will consider two types of constraints, **direct constraints** and **abstract constraints**. The above model has 5 direct constraints but only 1 abstract constraint. In general, we measure the **complexity** of a model in the number of abstract variables and constraints, and the **size** of the model is measured in the number of direct variables and direct constraints.

The corresponding Julia/JuMP model can now be written as:

```
IC = Model(HiGHS.Optimizer)

@variable(IC,x[c=1:C] >=0 )

@objective(IC, Max, sum( Profit[c]*x[c] for c=1:C)

@constraint(IC, [p=1:P],
sum( RecourseUsage[p,c] * x[c] for c=1:C) <= Capacity[p])
```

```
9
10 optimize!(IC)
```

Notice that the program actually becomes smaller. Also there is a one-to-one correspondence between the individual elements.

- The definition of the variable $x_c \geq \forall\ c$ is now given on one line, line 3:

```
@variable(IC,x[c=1:C] >=0 )
```

- The $\sum$ operator, over an index of a set, is used both in the objective function and in the constraint:
  - The objective function, line 5, $\sum_{c \in C} Profit_c \cdot x_c$ is written as:

```
@objective(IC, Max, sum( Profit[c]*x[c] for c=1:C))
```

  - The constraint: $\sum_{c \in C} RecourseUsage_{p,c} \cdot x_c \leq Capacity_p \forall p \in P$ is written in 7 and 8 as:

```
@constraint(IC, [p=1:P],
sum( RecourseUsage[p,c] * x[c] for c=1:C) <= Capacity[p])
```

In line 1, the data-structure in which the variables, objective function, and constraints are contained is constructed, named IC. Here the solver needed to solve the model is also defined.

```
IC = Model(HiGHS.Optimizer)
```

When the model is considered to be complete, it is transferred to the solver, which finds the optimal solution to the model. This happens in line 10:

```
optimize!(IC)
```

When the optimize! command is given, the full model is transferred to the solver, HiGHS in this case.

The Model construct, and the different model structures @variable, @objective, and optimize! are part of the JuMP package.

Setting up the data-structures Profit, Capacity, and RecourseUsage in Julia is described in Chapter 9. Below we present full Julia/JuMP program with the correct numbers and data-sets for the Incredible Chairs 2 assignment. This format is used in all of the assignments in the book.

Standard Incredible Chairs 2 model version:

```
#************************************************************************
# Incrediblex Chairs 2 assignment, Simple LP
using JuMP
using HiGHS
#************************************************************************

#************************************************************************
# Data
Chairs=["A", "B", "C", "D", "E", "F", "G", "H", "I", "J"]
C=length(Chairs)
ProductionLines=[1, 2, 3, 4, 5]
P=length(ProductionLines)

Profit=[6, 5, 9, 5, 6, 3, 4, 7, 4, 3] # profit for each chair
Capacity=[47, 19, 36, 13, 46] # production line capacity
RecourseUsage=[ # RecourseUsage[p,c] res use for chair c on prod. line p
6  4 2  3 1 10 2 9  3 5;
5  6 1  1 7  2 9 1  8 6;
8 10 7  2 9  6 9 6  5 6;
8  4 8 10 5  4 1 5  3 5;
1  4 7  2 4  1 2 3 10 1]
#************************************************************************

#************************************************************************
# Model
IC2 = Model(HiGHS.Optimizer)

@variable(IC2,x[c=1:C]>=0) # x[c] is the number of chairs produced

# Objective is to maximize profig
@objective(IC2, Max, sum( Profit[c]*x[c] for c=1:C ) )

# Constraints ensuring production line capacity is not exceeded
@constraint(IC2, [p=1:P],
            sum( RecourseUsage[p,c]*x[c] for c=1:C) <= Capacity[p]
            )
#************************************************************************
```

```
38
39 ##########################################################################
40 # Solve
41 optimize!(IC2)
42 println("Termination status: $(termination_status(IC2))")
43 ##########################################################################
44
45 ##########################################################################
46 if termination_status(IC2) == MOI.OPTIMAL
47     println("Optimal objective value: \$(objective_value(IC2))")
48     for c=1:C
49         if value(x[c])>0.001 # only print the chairs actually produced
50             println("No of chairs of type ", Chairs[c], " produced: ",
                ↪  value(x[c]))
51         end
52     end
53 else
54     # model is in-feasible or un-bounded
55     println("No optimal solution available")
56 end
57 ##########################################################################
```

The whole Julia/JuMP program consists of 5 overall parts, divided by the comment lines of asterisk. # is used for comments. The program parts are:

- The initial section, 1-5. Here the program is named and the Using keyword is used to load the different packages.
- The data section, 7-22. Here the data needed for the model is defined. For at description of how to define data, Chapter 9
- The model section, 24-37. Here the decision variables, objective function and constraints are defined.
- The solve section, 39-43. Here the model is transferred to the solver and solved.
- The result section, 45-57. The results from the solver are printed.

3.3 Good modelling practices and debugging

There are a number of recommended practices in the above example:

- Use comments extensively; it makes the programs more readable. The character # makes all text afterwards on that line a comment, which is shown in green above.
- Given a **set** of items, define them as a one-dimensional vector (Chairs line 9) **and** define the length of the vector, as an upper-case letter (C line 10).
- Choose your index names carefully! As an example, c for chairs and C for the number of chairs. This is particularly important since there is no type-checking of sets in Julia/JumP; The instruction RecourseUsage[c,p] is a valid instruction but will lead to an error, it should have been RecourseUsage[p,c].

3.3.1 Debugging

Errors can be one source of frustration when implementing mathematical models. It is important to have strategies to resolve them when they do occur. To provide an example of how one might resolve an error, we deliberately introduce a programming error in the above program and change line 41 to:

```
optimize!(ICC)
```

When calling Julia, we get the following error report:

```
ERROR: LoadError: UndefVarError: ICC not defined
Stacktrace:
 [1] top-level scope
   @ ~/Desktop/IncredibleChairs.jl:41
in expression starting at /Users/thomasstidsen/Desktop/IncredibleChairs.jl:37
```

Error messages may not be very descriptive at times, and very often the best clue is to look for the line number of the error (here 41). The programs that are considered in this book are all of a relatively limited size, usually less than 200 lines of code. However, constraints that conflict with each other can lead to errors which are very difficult to spot.

Assuming that there are no Julia programming errors (as above), the objective of the model can be:

- An optimal and correct solution. We check if an optimal solution is found in line 46. If yes, he solution here is printed out 45-57. Notice use of the two JuMP functions:

1. objective_value(IC2)) : After optimization, if there is an optimal result, we can obtain the optimal value using this function on the model.
2. value(x[c]) : After optimization, if there is an optimal result, we can obtain the optimal value, i.e. the optimal number of chairs which should be produced, using this function

- An optimal but incorrect solution: In all of the assignments in this book, the optimal objective value is given. If your model does **not** give this value, then there is definitely an error in your model. If your model **does** get the same optimal value, then your model **may** be correct, but this is not a guarantee.
- Unbounded: If the test in line 46 evaluates to false, the model may be unbounded. This means that the objective tends to $+\infty$ (for a maximization problem) or $-\infty$ (for a minimization problem). In both cases, there is an error in your model (this is not a lost opportunity to earn an infinite amount of money). Often this error comes down to two possibilities: You have forgotten to set the domain of your variables, e.g. $x_i \geq 0$ or you have inadvertently omitted a constraint that limits values of one or more of your decision variables. In the above example, if we add an extra chair, chair no. 11, and define the resource usage to be 0 for all production lines (e.g. it could be produced on a new line we forget to include), and there is a positive profit value, i.e. Profig[11]>0, **then** the solver will try to produce an infinite number of chairs of type 11.
- Infeasible: This means that there is no feasible solution to the model. This can happen; however, in this book all assignments have a feasible, optimal solution. In the above example, if just one of the capacity values take a negative value e.g. Capacity[1]=-47, the whole model becomes infeasible. The simplest way to debug this case is to insert sets of constraints one-by-one to see at which point feasibility of the problem is lost.

3.4 Conditional sums and constraints

Sometimes it is necessary to make conditional constraints or to use sum over a subset of elements in the objective function and/or constraints. This is easy to implement in Julia/JuMP:

- **Conditional constraint:** $\sum_k x_{i,j,k} <= 1 \ \forall \ i, j | i \neq j$

```
@constraint(M, [i=1:I,j=1:J;i!=j], sum( x[i,j,k] for k=1:K) <= 1)
```

  Notice that here the ";" defines the beginning of the condition.

- **Conditional sum:** $\sum_{i,j|i \neq j} x_{i,j}$

```
sum( x[i,j] for i=1:I,j=1:J if i !=j )
```

Notice thate here "if" defines the beginning of the condition.

This kind of conditional statement is necessary in some of the assignments.

3.5 Balance constraints

A very common constraint type used in LP and MIP models is the so-called **balance constraint**. Such a constraint connects two different sets of variables and ensures that the value of their respective sums is exactly the same, i.e., balanced.

Three examples of balanced constraints that you will encounter throughout the exercises include the following:

- *Product flows*: A storage facility where products are first transported from factories into the storage facility and which are then subsequently transported from the storage facility to the customers. Since there is no production at the storage facility and nor is it possible to store products long term, the quantity of product is delivered from the different factories to a storage facility should match precisely the quantity of product delivered from the storage facility to the customers.

- *Inventory management of a certain product*: For a given time horizon, and an initial storage level of the product, the quantity of product on hand at the end of the time horizon must be the initial storage level adjusted by any new production (increasing the quantity on hand) and the amount of product that is sold or released within the time horizon (decreasing the quantity on hand).

- *Flows within a network*: As an example, consider intersections in a road network. The cars driving into any intersections must leave it again.

A good example that you will meet in a later assignment is the inventory management case. This requires one to relate a given timeslot (index) with its previous timeslot (index) and induces boundary certain boundary cases. Consider what is the timeslot before the the first timeslot. In Julia, the in-line if statement can be used. `(A ? B : C)` is an if statement. It evaluates whether `A` is true or false. If `A` is true, then it uses `B` and if `A` is false, then it uses `C`. Therfore, a statement like `(t>1 ? <normal case> : <what to do if this is first time-slot>)` can be used.

3.6 Relaxation and restriction of linear programs

Consider an LP model. By changing its set of constraints, one can either **relax** the model or **constrain** the model further.

Relaxation: If one (or more) of the constraints is removed from the model, then the model has been *relaxed*. The resulting model is termed a *relaxation*. Relaxations lead to optimal objective values that are at least as good as the original model (i.e. an increased value if maximizing and a decreased value if minimizing). The reason for this is that all of the previous solutions are still feasible, but there might also be new solutions that are now feasible due to the omitted constraints.

Restriction: If one (or more) constraint(s) is added to the model, then the model has fewer feasible solutions. Constraining a model more leads to (possibly) worse optimal objective values (i.e., a decreased value if maximizing and an increased value if minimizing). The reason for this is that some of the previously feasible solutions may now be infeasible given the newly added constraints.

3.7 Solver parameters

It is possible to transfer parameters to the solver via Julia/JuMP. This is often not so important for LP models since these are often solved quickly. When MIP models are solved, however, it may be interesting to set, e.g., a time limit or an acceptable optimality gap. Below we provide examples of how to set the values of some useful solver parameters. In the examples, M refers to the model name.

- Set a time limit for the solver. This may mean that the solver is forced to terminate after a specific time. The solver may not find the optimal solution or even find a feasible solution. Below, we demonstrate how the the time limit is set to 300 sec.

  ```
  set_time_limit_sec(M, 300.0)
  ```

- The printout from to solver suppressed with the set_silent function:

  ```
  set_silent(M)
  ```

- Set an acceptable gap, i.e. an acceptable relative gap between the best possible objective value and the current best value. This option becomes critical when working with MIP. This parameter depends on which solver is used:

- Setting a gap of 10 % in HiGHS:

  ```
  set_optimizer_attribute(M, "mip_rel_gap", 0.10)
  ```

- Setting a gap of 10 % in Gurobi:

  ```
  set_optimizer_attribute(M, "MIPGap", 0.10)
  ```

- Setting a gap of 10 % in Cplex:

  ```
  set_optimizer_attribute(M, "CPXPARAM_TimeLimit, 0.10)
  ```

3.8 Julia and JuMP structure

It is important to understand the differences and the relationships between Julia, JuMP, and the solver. In this section, we will briefly describe this.

3.8.1 Julia

Julia is a fully-fledged programming language that is used in many different application areas. In this book, we will only use a small subset of the Julia constructs.

3.8.2 JuMP

JuMP is a package for Julia and is specialized for creating Mathematical Programming models. In this book, we will only consider LP models and MIP models. JuMP can, however, be used for a number of other model types. JuMP extends Julia with some new constructs, created as macros. Consider the following example taken from Incredible Chairs, see Chapter 3.1.

- The model statement:

  ```
  IC = Model(HiGHS.Optimizer)
  ```

  The model statement is used to define model.

- The variable statement:

  ```
  @variable(IC,xA>=0)
  ```

  The variable statement is used to define a variable for a given model, in this case IC.

- The objective statement:

  ```
  @objective(IC, Max, 4*xA+6*xB)
  ```

  The objective statement is used to define an objective function for a given model, in this case IC.

- The constraint statement:

  ```
  @constraint(IC, 4*xA+3*xB <= 36)}
  ```

  The constraint statement is used to define one (or more constraints) for a mathematical programming model, in this case IC.

- The optimize statement:

  ```
  optimize!(IC)
  ```

  The optimize statement is used to transfer the model to the **solver**.

- The value statement:

  ```
  value(xA)
  ```

  This function is used to retrieve the value of a JuMP variable in the solution found by the solver. This function can only be used **after** the optimize! statement.

There are a number of important details:

- The first statement is always the model statement.
- The order of variable, constraint and objective statements does not matter. In this book, we will always first write the variables, then the objective function and finally the constraints
- There can be multiple variables, of different names, and multiple different constraints, but only one objective function. If several objective statements are issued, only the last before the optimize statement is used.
- The optimize statement builds the model in a solver-readable form and transfer the control to the solver.
- The variables of a JuMP model do **not** hold a value until after the optimize statement, The solver assigns the values, which can then be retrieved using the value statement. For this reason, the JuMP variables cannot be used in Julia constructs prior to the optimize statement.

3.8.3 Solver

One of the major advantages of using JuMP is that one can easily toggle between different solvers for the same model. The only changes necessary are in using statement at the top of the program and also in the model construction statement. JuMP supports a number of solvers for both LP models and MIP models, see the list of solvers we recommend in Section 2.3.

All models in this book can be solved in reasonable time using the opensource HiGHS solver. The two leading solver commercial solvers Gurobi and Cplex are much faster, in particular for MIP models, but they are relatively expensive. Academics and students can usually get access to an academic license for the solvers, but this requires a separate installation. The opensource solvers are, however, very simple to install in Julia, using the package system.

3.8.4 Structure

We are now ready to briefly describe the interactions of a model in Julia/JuMP with the solver, see Figure 3.1. The lowest level, i.e., the green part, corresponds to the Julia code part. When the program is executed, the initial part and any data will be initialized in the Julia program. This only affects the Julia part. The model is then built using the JuMP package. This is essentially an interface between the Julia program and the solvers (of which three are indicated in the figure). The JuMP package provides the model function and the @variable, @objective, and @constraint commands. These commands are used to build the LP model or MIP model. When the optimize statement is executed, the constructed JuMP model is transferred to the solver, which will then start to solve the model. When the solver terminates, the optimal objective value and corresponding solution is available via JuMP and can be queried from the Julia part using the respective functions objective_value function() and the value().

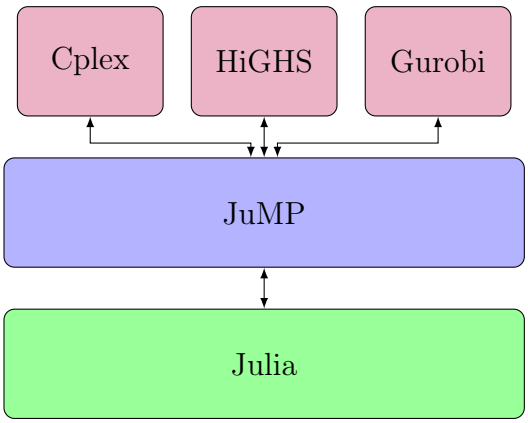


Figure 3.1: Julia/JuMP and Solver Structure

An important point is that the values of the JuMP variables, defined with the @variable statement, **cannot be used as variables in the Julia program !**. This is because, the value is set **only** by the solver and it can afterwards only be queried, not altered, using the value() function. We illustrate this with two deliberate errors in the Julia/JuMP program below (both will lead to errors in the compilation):

- In line 10, we attempt to make an in-line conditional statement in a constraint using the value of the JuMP variable xA. This is wrong for two reasons:
  - The xA JuMP variable does not have a value before the optimize!() statement.
  - Even if it had a value, the value() function is needed to extract this value
- In line 13, the value of the xA JuMP variable is used in an if statement, which is an error for the same reasons.

In line 21, the value() statement is used correctly to obtain the optimal value of the xA JuMP variable.

```
1 using JuMP
2 using HiGHS
3 IC = Model(HiGHS.Optimizer)
4
5 @variable(IC,xA>=0)
6 @variable(IC,xB>=0)
7 @objective(IC, Max, 4*xA+6*xB)
8 @constraint(IC, 2*xA <= 14)
9 @constraint(IC,3*xB <= 15)
```

```
@constraint(IC,4*xA+3*xB <= 36 + $(xA>2 ? 10 : 0 ) ) # <<<<<< error
print(IC)

if xA>3 # <<<<<< error
    println("High production of chair A")
end

optimize!(IC)
println("Termination status: $(termination_status(IC))")
if termination_status(IC) == MOI.OPTIMAL
    println("Optimal objective value: $(objective_value(IC))")
    println("xA: ",value(xA)) # <<<<<< correct
    println("xB: ",value(xB))
else
    println("No optimal solution available")
end
```

4 Linear Programming problems

This chapter is the first exercise chapter of the book. All the assignments in this chapter focus on LP.

4.1 Jewelry Production

Your aunt runs a small jewelry production company that produces relatively cheap necklaces (five different types), primarily for tourists. The profit for each necklace, price with material costs subtracted, is given in Table 4.1.

	Necklace type				
	1	2	3	4	5
Profit (€)	50	45	85	60	55

Table 4.1: Profit per necklace type (€)

Three machines are used to produce different jewelry components, which are then assembled by hand by two employees. Your aunt handles the machines herself; they are automatic but require some training to use. As they are automatic, all machines can run at the same. To produce the components for each necklace different amounts of time are required on each of the three machines. The machine time needed (in minutes) is given in Table 4.2. It is not necessary to produce components in a specific order.

	Necklace type				
Machine	1	2	3	4	5
1	7	0	0	9	0
2	5	7	11	0	5
3	0	3	8	15	3

Table 4.2: The machine time needed for each necklace (minutes)

Additionally, each necklace must be assembled by one worker, and this takes some time (also measured in minutes). These times are given in Table 4.3. A full work day, both for the assembly and for each machine, is 7.5 hours.

	Necklace type				
	1	2	3	4	5
Assembly time	12	3	11	9	6

Table 4.3: Necklace assembly time (in minutes)

Assignment 1.1

Formulate an LP that can be used to maximize the daily profit. You must determine the number of each type of necklace to produce, while taking into consideration the available machine and assembly hours.

Optimal objective value for Product Mix: €5896.15.

This is a simple example of a so-called *product mix* problem, a common application of LP.

Your aunt realize that she cannot sell all of the necklaces she makes, but estimates an average demand for each type. These values are given in Table 4.4

	Necklace type				
	1	2	3	4	5
Average Demand	25	10	12	15	60

Table 4.4: Average demands

Assignment 1.2

Find a revised production plan that maximizes the daily profit. The production of the necklace types should not exceed the average demand values. Before you implement this constraint, try to reason if the additional constraints are going to increase or decrease the profit?

Optimal objective value for jewellery production: €5643.18.

Your aunt considers whether to hire another assembly worker. Is this a good idea ?

4.2 Micro brewery

You and a friend want to set up a microbrewery, both, to satisfy your own demand, but also to supply some nearby cafes with beer. The deal is that the cafes will get the beer relatively cheaply, but they will have to buy a certain amount (in liters) each month. The total collective monthly demand is given in Table 4.5:

	Months											
	jan	feb	mar	apr	may	jun	jul	aug	sep	oct	nov	dec
Demand (L)	15	30	25	55	75	115	190	210	105	65	20	20

Table 4.5: Monthly demand for beer each month (in liters)

You set up the brewing system in your own room (you live at home with your parents) but can only brew at most 120 liters per month. This is a problem, given the demand, so you convince your mother that it is alright to store crates of beer in the basement. Her demand (to avoid excessive storage) is a storage cost of 1 € per liter of beer stored each month. Furthermore, you can store at most 200 liters of beer in the basement.

Assignment 2.1

Formulate an LP that can determine the optimal beer production and storage strategy for one whole year, minimizing the storage costs. Assume you start with no beer in storage in January.

Hint: You need a beer production variable for each month and a storage variable for each month. The storage variable should reflect what is in storage *at the end of the month*. Then, you can make a *balance constraint*, relating what is in storage at the end of a month, with what was in storage at the end of the previous month. The previous month is the current month minus 1. *However*, if it is the *first* month (jan), then there is no storage. If your storage variable is $y[m]$ (where m is the month index), you can add the previous month in Julia with the following code:

```
(m>1 ?  y[m-1] :  0)
```

In general, this code `(A ? B : C)` is an if statement. It evaluates whether `A` is true or false. If `A` is true then it uses `B` and if `A` is false then it uses `C`.

Assignment 2.2

Implement your LP model in Julia/JuMP.

Optimal objective value for micro brewery broblem: €560.0

4.3 Blending [1]

Cooking oil is manufactured by refining raw oils and blending them together. The five different raw oils come in two categories, Vegetable and Non-vegetable. An overview is given in Table 4.6.

Vegetable oils:	VEG 1
	VEG 2
Non-vegetable oils:	OIL 1
	OIL 2
	OIL 3

Table 4.6: The different oils

Vegetable oils and non-vegetable oils require different production lines for refining before they can be blended into the final product. In any month, it is not possible to refine more than 200 tons of vegetable oil and more than 250 tons of non-vegetable oils. There is no loss of weight in the refining process, and the cost of refining may be ignored.

There is a technological restriction of hardness in the final product. The final product must have a hardness value between 3 and 6, both values included, in the units used for measuring hardness. It is assumed that hardness blends linearly, i.e. if you use 2 tons of OIL 1 (hardness 2.0) and 3 tons of VEG2 (hardness 6.1), you get 5 tons of final product with a hardness of: (2*2.0+3*6.1)/(2+3)=4.46. The costs (per ton) and hardness of the raw oils are given in Table 4.7:

	VEG 1	VEG 2	OIL 1	OIL 2	OIL 3
Cost (£/ton)	110	120	130	110	115
Hardness	8.8	6.1	2.0	4.2	5.0

Table 4.7: Cost and hardness of the oils

The final product sells for £150 per ton.

Assignment 3.1

How should the cooking oil be made in order to maximize the net profit? To find out, make a Julia/JuMP model to plan the production

[1] Taken from Williams: "Model Building in Mathematical Programming"

Optimal objective value for Blending Problem: £17592.59

Tip: If the solver complains that the problem is *non-linear*, you have a problem: The model has to be linear (variables may be subtracted/added and multiplied with parameters but not e.g. multiplied/divided by each other). Can you change the constraint, such that it becomes linear but still work in the same way?

This is another very common type of application of linear programming although, of course, practical problems will generally be much bigger.

4.4 Blending 2 [3]

This is an extension of the Blending assignment. Instead of planning the production for just one month, production must now be planned for the six months from January to June (including both months). For each month of production, the hardness requirement of the finished product has to be satisfied, as before. However, the cost for buying the raw oils now varies between months. Raw oil may be purchased for immediate refinement and used in that month's final product or stored (unrefined) to be used later. Prices for the raw oils in the different months are given below (in £/ton):

	VEG 1	VEG 2	OIL 1	OIL 2	OIL 3
January	110	120	130	110	115
February	130	130	110	90	115
March	110	140	130	100	95
April	120	110	120	120	125
May	100	120	150	110	105
June	90	100	140	80	135

Table 4.8: Oil prices (£/ton)

The final product (still) sells at £150 per ton, and the refinement capacities are the same, for each month. It is possible to store up to 1000 tons of each raw oil for later use. The cost of storage for both vegetable or non-vegetable oil is £5 per ton per month. The final product cannot be stored. Nor can refined oils be stored. At the beginning of January there are 500 tons of each type of raw oil in storage. It is required that these stocks will also exist at the end of June.

Assignment 4.1

What buying and manufacturing policy should be adopted in order to maximize profit? Formulate this as an LP and implement the model in Julia/JuMP.

Optimal objective value for Blending 2: £107842.59

If you get £82750.00, take a look at your refinement capacity constraints, do they limit the right variables?

[3] Taken from Williams: "Model Building in Mathematical Programming"

4.5 Factory Planning [4]

An engineering factory makes seven products (PROD 1 to PROD 7) that each require certain manufacturing processes (grinding, vertical drilling, horizontal drilling, boring, or planing). The factory has four grinders, two vertical drills, three horizontal drills, one borer, and one planer. Each product yields a certain amount of profit (selling price minus the cost of raw materials pr unit). Additionally, each product requires a certain amount of process time with each type of machine. Profits and process times are shown in Table 4.9. A dash indicates that the product does not require processing with that machine.

		PROD						
		1	2	3	4	5	6	7
	Profit (£/unit)	10	6	8	4	11	9	3
Process Time (hours)	Grinding	0.50	0.70	-	-	0.30	0.20	0.50
	Vertical drilling	0.10	0.20	-	0.30	-	0.60	-
	Horizontal drilling	0.20	-	0.80	-	-	-	0.60
	Boring	0.05	0.03	-	0.070	0.10	-	0.08
	Planing	-	-	0.01	-	0.05	-	0.05

Table 4.9: Product details

In the present month (January), and the five subsequent months, certain machines will be down for maintenance and cannot be used that month. A summary of the machines that will undergo maintenance is given in Table 4.10.

Month	Maintenance
January	1 grinder
Februrary	2 horizontal drills
March	1 borer
April	1 vertical drill
May	1 grinder and 1 vertical drill
June	1 planer and 1 horizontal drill

Table 4.10: Machine maintenance

The numbers of units of each product that can be sold in each month is limited by the market demand and are given in Table 4.11. Furthermore, it is possible to store up

[4] Taken from Williams: "Model Building in Mathematical Programming"

to 100 units of each product at a cost of £0.5 per unit per month. Products must be finished within one month, i.e. it is not possible to store half finished products between months, only finished products that have been processed fully on all machines they require. There are currently no units of any item in stock, but the factory would like to have 50 units of each type of product on hand at the end of June. The factory works a 6 day week with two shifts of 8 hours each day. You may assume that each month consists of 24 working days. No sequencing considerations need to be given when processing the products on the machines i.e. it is only necessary to consider the necessary time on the various machines not which order the processing takes place in.

	PROD						
	1	2	3	4	5	6	7
January	500	1000	300	300	800	200	100
February	600	500	200	0	400	300	150
March	300	600	0	0	500	400	100
April	200	300	400	500	200	0	100
May	0	100	500	100	1000	300	0
June	500	500	100	300	1100	500	60

Table 4.11: Market limitations

Assignment 5.1

Formulate an LP that can be used to determine when and what the factory should produce in order to maximize their profit. Implement your model in Julia/JuMP to find the optimal solution.

Optimal objective value for the Factory Planning Problem: £93715.17 (ultimo storage)
Optimal objective value for the Factory Planning Problem: £93890.17 (primo storage)

Hint: Think carefully about when you "count" your storage, it matters if you do it in the beginning of the month (*primo*) or at the end of the month (*ultimo*).

Assignment 5.2

Estimate the value of acquiring new machines.

4.6 Project Scheduling

Your father is unhappy with the tool shed in the back garden of his house, see the picture. He wants a new one and has found a do-it-yourself kit tool shed from a hardware store. There is, however, a significant amount of work which has to be done. He splits the total project into tasks, some of which can be done by you, others must be done by specialists (electricians, plumbers etc.). He asks all contractors to commit to a day usage for the task. He offers you the first and last task and would like you to be in charge of the project. Given the list below, you need to make a *project plan*, i.e. when should each task be performed. To keep things simple, we will measure time in days. Notice that tasks can be carried out simultaneously, but some tasks require that other tasks have been completed before they can be started. As an example, the roof can only be installed after the walls have been erected.

Figure 4.1: The tool shed to be replaced

The list of all tasks to be completed is as follows:

1. Remove Stuff: Move the tools and other stuff out of the current shed and temporarily store them (duration 1 day)

2. Shed Demolition: Knock down the old shed (after Remove Stuff, duration 3 days)

3. Pipe Work: Install plumbing, there will be running water in the new shed (after Shed Demolition, duration 5 days)

4. Electric Work: There is electricity for tools and lights in the new shed (after Shed Demolition, duration 3 days)
5. Flagstones: The area around the shed should be covered in flagstones (after Pipe Work and Electric Work, duration 7 days)
6. Shed Foundations: Set the foundations for the new shed (after Pipe Work and Electric Work, duration 4 days)
7. Wood Frames: Erect the wooden walls of the shed (after Shed Foundations, duration 3 days)
8. Painting: Paint the walls (after Wood Frames, duration 1 day)
9. Roofing: Build the roof (after Wood Frames, duration 1 day)
10. Roofing Felt: Put roofing felt on the roof (after Roofing, duration 1 day)
11. Interior Installations: Install racks, lights etc. (after Roofing, duration 3 days)
12. Insert Stuff: Get the stuff out of the temporary storage and store them in the new shed (after Roofing, duration 1 day)
13. Hand over: Present the new shed to your father (After Flagstones, Painting, Roofing felt, Interior installations and Insert stuff, duration 0 days)

Assignment 6.1

Formulate the problem of minimizing the number of days needed to complete the shed as an LP. You may want to consider achieving this by minimizing the start time of the final task, "Hand over". The plan should take into account precedence restrictions of tasks. Implement and solve the model in Julia/JuMP.

Optimal objective value for the Shed Project Scheduling : 20.0 days

Hint: You can use conditional statements to only create constraints on the start time of tasks for which other tasks need to be fully completed before they may be started. See Section 3.4 for more on conditional constraints.

Early finish
Your father is not happy about the project plan, he needs the tool shed to be finished earlier. Looking at the plan, you contact all the contractors, who use more than one day for the job, and ask how much they can speed up the work, and how much overtime pay they need per speed-up day. This gives the information below.

- Shed Demolition: Can be reduced by up to 1 day at an overtime pay of 500 per day
- Pipe Work: Can be reduced by up to 3 days at an overtime pay of 2000 per day
- Electric Work: Can be reduced by up to 1 day at an overtime pay of 4000 per day
- Flag Stones: Can be reduced by up to 4 days at an overtime pay of 1000 per day
- Shed Foundations: Can be reduced by up to 2 days at an overtime pay of 1000 per day
- Wood Frames: Can be reduced by up to 2 days at an overtime pay of 1500 per day
- Interior Installations: Can be reduced 2 days at an overtime pay of 1500 per day

You realize that there is a problem. The completion time depends on the budget available for overtime, and the question your father has to answer is: How much is he prepared to pay for an earlier finish? Before you return to your father you first calculate how quickly the tool shed can be completed if there is an unlimited budget. For comparison, you also calculate the earliest completion times if you have a budget of 5000 and if you have a a budget of 10000.

Assignment 6.2

Implement the Shed Project Scheduling in Julia/JuMP with the possibility to speed up tasks and minimize the number of days before hand over, taking into account an unlimited budget, a 5000 budget and a 10000 budget.

Optimal solution for the Shed Project Scheduling, unlimited budget: 10.0 days
Optimal solution for the Shed Project Scheduling, budget of 5000 : 15.33 days
Optimal solution for the Shed Project Scheduling, budget of 10000 : 12.80 days

5 Mixed Integer Programming

LP models are very useful, but there are many situations where the planning problem in question cannot be modelled sufficiently precisely. In many problems fractionality of the variables is problematic. Consider the Incredible Chairs problem again, now Incredible Chairs III: Since it is chairs that are produced, they need to be produced in **whole** numbers. Updating the model to achieve this in Julial/JuMP is simple (last line of model):

$$
\begin{array}{lrcrcl}
\texttt{Maximize.} & 4 \cdot x_A & + & 6 \cdot x_B & & \\
\texttt{Subject to :} & & & & & \\
 & 2 \cdot x_A & & & \leq & 14 \\
 & & & 3 \cdot x_B & \leq & 15 \\
 & 4 \cdot x_A & + & 3 \cdot x_B & \leq & 36 \\
 & x_A, x_B \in Z^+ & & & &
\end{array}
$$

This is a Mixed Integer Programming (MIP) model where the variables are allowed to be restricted to only allowing **integer** values, as described above in the **domain constraints**. The below program will deliver the optimal **integer** solution. Only two things have been changed:

- In the definition of the variables, the variable is now declared to be integer, line 5 and line 6:

```
@variable(IC,xA>=0,Int)

@variable(IC,xB>=0,Int)
```

- When retrieving the value of the variable found by the solver, using the value() function, we **round** the resulting variable to become an integer type, line 16 and line 17:

```
println("xA: ",round(Int64,value(xA)))

println("xB: ",round(Int64,value(xB)))
```

Compact IC model version:

```
using JuMP
using HiGHS
IC = Model(HiGHS.Optimizer)

@variable(IC,xA>=0,Int)
@variable(IC,xB>=0,Int)
@objective(IC, Max, 4*xA+6*xB)
@constraint(IC, 2*xA <= 14)
@constraint(IC,3*xB <= 15)
@constraint(IC,4*xA+3*xB <= 36)
print(IC)
optimize!(IC)
println("Termination status: $(termination_status(IC))")
if termination_status(IC) == MOI.OPTIMAL
    println("Optimal objective value: $(objective_value(IC))")
    println("xA: ",round(Int64,value(xA)))
    println("xB: ",round(Int64,value(xB)))
else
    println("No optimal solution available")
end
```

From a modelling point of view, these changes are small and almost trivial. However, three important points need to be made, which are elaborated upon in the below sections:

1. The amount of important problems which can be better modelled using MIP models is **vast**. In the next section, we will briefly present a number of integer modelling tricks.

2. The required solution time of the solver may become critical: Finding the optimal solution for a MIP model becomes a much more challenging computational problem than for an LP model.

3. The type of algorithm necessary to find the optimal solution changes. To better understand the increased optimization complexity, we give a very brief introduction to the classic Branch & Bound algorithm.

5.1 Modelling power

At first glance, the addition of integer requirements for the variables does not seem like such a big change. However, many problems are **inherently** integer:

- *Chair production, or any type of production of integer amounts* - It is hard to sell fractional chairs.
- *Vehicle routing, e.g. trucks, ships, trains* - a fractional ship cannot sail.
- *Manpower planning* - planning with fractional workers is not a good option.
- *Timetabling* - the classes could be divided up, but what about the teacher?

These are just a few examples. It turns out that there are many cases where one or more variables needs to be integer ($x \in Z$) or binary ($x \in \{0, 1\}$).

Generating a good integer solution from the Incredible chairs model is not complicated. The optimal value for Incredible chairs is 51 with the values $x_A = 5.25$ and $x_B = 5$. Because all the coefficients of x_A are positive in the constraints and all the constraints are less-equal $\leq$, x_A can simply be rounded down to $x_A = 5$, achieving an integer solution. This reduce the objective value to 50. But is this solution optimal? In the Incredible Chairs 3 assignment this is the topic.

In general, just rounding fractional variables is **not** guaranteed to reach an optimal solution, perhaps not even a feasible solution. In the Incredible Chairs case, the integer solution objective went from 51 to 50. In other cases, the changes in the objective can be far greater: Suppose a container ship has to be routed to pick up containers. If 3 containers have to be picked up from a port by a container ship with a capacity of 1000, an LP model will assign a $\frac{3}{1000}$ ship to do the job. If what is minimized is cost, the objective can increase substantially by increasing the value from $\frac{3}{1000}$ to 1.

An important special case of integer variables is **binary** variables i.e. $x \in \{0, 1\}$. This can model yes/no decisions, usually interpreting 1 as yes and 0 as no. A set containing C binary variable can easily be defined in Julia/JuMP as:

```
@variable(IC,x[c=1:C],Bin)
```

where IC is the name of the mathematical model. Notice that since a binary variable can **only** take the values 0 or 1, adding lower and upper bounds is not necessary. For integer variables however, upper bounds can be added in exactly the same way as for continuous variables.

```
@variable(IC,x[c=1:C],Int)
@variable(IC,x[c=1:C] >= 0,Int)
@variable(IC,0 <= x[c=1:C] <= 10,Int)
```

5.1.1 Integer modelling tricks

There are a number of standard modelling tricks which can be utilized when modelling with integer or binary decision variables. Here we briefly mention the most important constructions, and point to the assignments where they are needed.

Fixed charge link: Often a resource like a ship, depot, truck or plant, is represented by a binary variable (is it needed or not) or how many do we need. The cost of using the resource is then 0 if it is not used, but even the most limited use incurs a fixed cost. Mathematically speaking, this can be stated as:

$$h(x) = \begin{cases} 0, & x = 0, \\ f + px, & x > 0. \end{cases}$$

If we assume that there is an upperbound on $x \leq B$, then this can be modelled using binary decision variables in the following way:

$$\begin{aligned} h(x) &= fy + px, \\ x &\leq By, \\ y &\in \{0, 1\}. \end{aligned}$$

In Julia this would look like this:

```
@variable(IC,x>=0)
@variable(IC,y,Bin)
@variable(IC,hx>=0)

@constraint(IC, x <= B * y)
@constraint(IC, hx == f * y + p * x) # the resulting function
```

Such a fixed charge link could, e.g., correspond to the cost of transporting x containers on a container ship: There is an additional cost, due to extra fuel needed for the extra weight of the container, but the main cost is probably due to the need for a full ship, i.e. the y cost. This type of modelling is needed in Chair Logistics 2.

If the resource can be increased in fixed amounts, then the only change needed is to change the definition of the resource variable from binary to integer. This could correspond to, e.g., a number of container ships.

Logic on binary variables
Often binary variables can be used to formulate logical relations between different binary variables. Usually, a binary variable $x = 1$ corresponds to true and $x = 0$ corresponds to false. With these definitions, we can easily define the various logic operators:

OR: Given two binary variables x and y, define the v binary variable as $x\ OR\ y$:

$$\begin{aligned} v &\geq x \\ v &\geq y \\ v &\leq x + y \end{aligned}$$

AND: Given two binary variables x and y, define the v binary variable as $x\ AND\ y$:

$$\begin{aligned} v &\geq x + y - 1 \\ v &\leq x \\ v &\leq y \end{aligned}$$

NOT: Given one binary variable x, define the v binary variable as $NOT\ x$:

$$v = 1 - x$$

Force selection: Given a set of i items, select at least 3 of these:

$$\sum_i x_i \geq 3$$

Limit selection: Given a set of i items, select at most 4 of these:

$$\sum_i x_i \leq 4$$

A classical mistake in modelling is to try to model logic by using JuMP variables as Julia variables, as discussed in Section 3.8.4. Since this is not possible, when logic in the model needs to modelled, it can be modeled as above. This is a requirement in the Startup Fund assignment 6.2.

Binary multiplication: All MIP model problems in this book can be modeled using **linear** models, i.e. where the variables are not multiplied. Given two binary variables x and y the multiplication of these can be formulated in a linear fashion, where the binary variable z is the result: (exactly as the x AND y operator)

$$
\begin{aligned}
z &\leq x \\
z &\leq y \\
z &\geq x + y - 1
\end{aligned}
$$

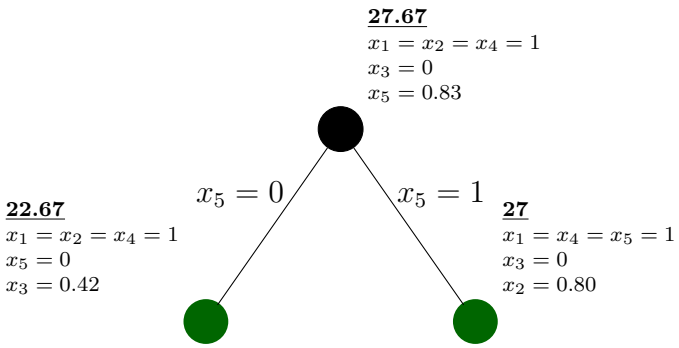
5.2 Solution algorithm: Branch & Bound

The focus in this book is on **modelling** LP and MIP models, not on the optimization algorithms needed to find the optimal solution to the models. In Chapter 8, we give references for further study, if you want to dive into this. It is, however, beneficial to have some intuition about the classical Branch & Bound algorithm for solving MIP models in order to better model different problems. In this section, we give a very simple and intuitive description of the basic approach.

The description of the algorithm is going to consider a very simple, small MIP model, the knapsack problem, see below:

$$
\begin{aligned}
&\texttt{Maximize.} && 7 \cdot x_1 + 10 \cdot x_2 + 4 \cdot x_3 + 4 \cdot x_4 + 8 \cdot x_5 && \\
&\texttt{Subject to :} && && \\
& && 3 \cdot x_1 + 5 \cdot x_2 + 12 \cdot x_3 + 2 \cdot x_4 + 6 \cdot x_5 && \leq \quad 15 \\
& && x_1, x_2, x_3, x_4, x_5 \in \{0, 1\} &&
\end{aligned}
$$

The knapsack problem contains 5 binary variables. In the Branch & Bound algorithm (B&B), first the LP version of the above problem, where the variables are **relaxed**, i.e. $x_1, x_2, x_3, x_4, x_5 \in [0, 1]$, is solved. This problem can easily be solved using an LP solver, reaching the optimal value 27.67, with the values: $x_1 = x_2 = x_4 = 1$ and $x_3 = 0$ and $x_5 = 0.83$. This solution is almost an integer solution, and the value 27.67 is a valid upper bound, i.e. we now know that **no** integer solutions with higher objective values exist. The B&B algorithm then selects the branching variable, i.e., selects one of the variables that is fractional and forces it to be either 0 or 1 by considering two new problems, where each of the new problems has an additional constraint. In one of the problems, $x_5 = 0$, and in the other $x_5 = 1$. This leads to two new solutions, shown as branches in a binary tree.

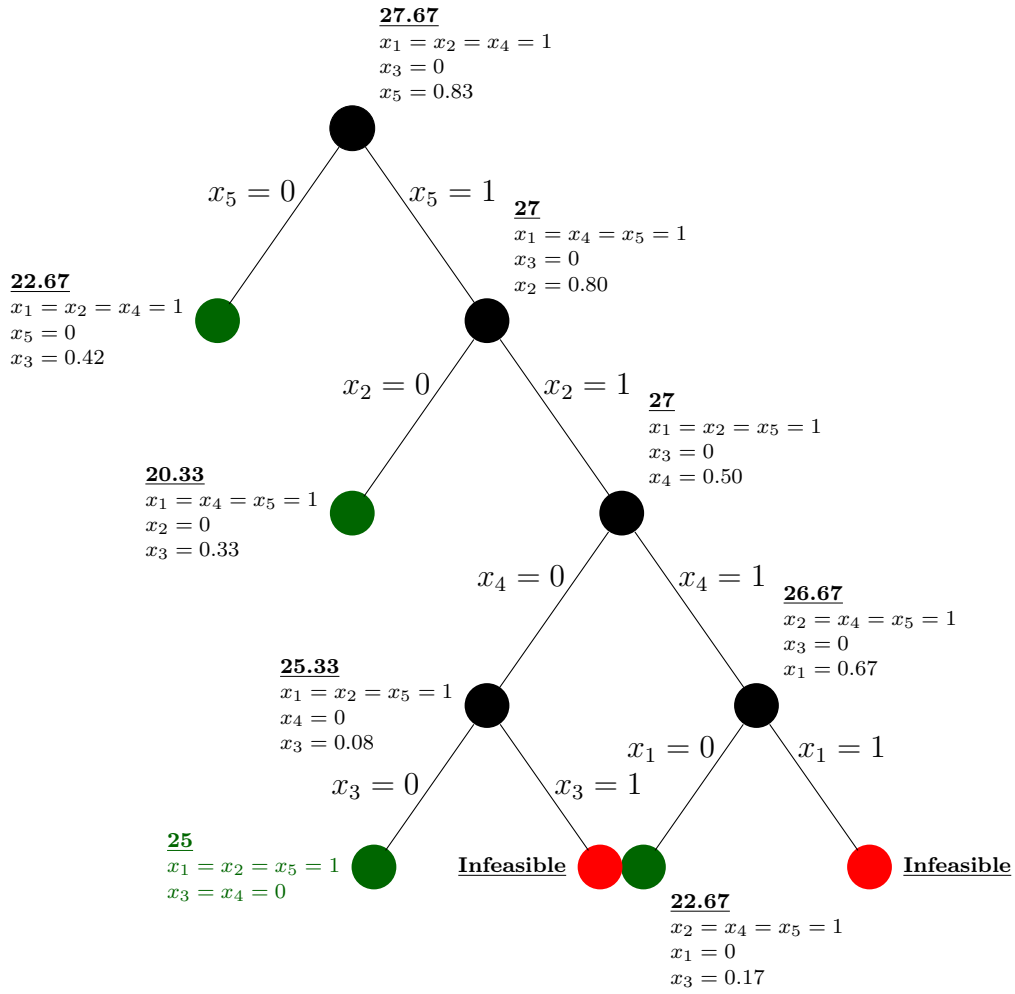


The black node is the so-called root solution. This is the LP relaxation we solved above and has an objective value of 27.67. The two new, green nodes reflect the impact of branching on variable x_5. The two problems are then solved giving solutions with objective values 22.67 ($x_5 = 0$) and 27 ($x_5 = 1$), respectively. The additional constraints are **restrictions** as mentioned in Section 3.6 and therefore the objective in both branches cannot have a higher value than 27.67. There are three possibilities for each of the new problems. These are:

- **The problem is infeasible:** It is not possible to find a (fractional or integer) solution, which is feasible. This means that any solution with more constraints will also be infeasible. Thus, we can **fathom** this node, i.e. not consider this node solution and any solution derived from it by branching. Fathoming is possible since it is guaranteed that no solution in the sub-tree of this node can improve the current best solution because all the solutions in the sub-tree will be infeasible.
- **The solution is integer:** This may be the optimal solution. If it is either the first solution found or better than the current best, then the solution is saved and termed the **incumbent**. It is the best integer solution found so far. All solutions in the sub-tree of this node can be ignored since they will be at best be as good as this solution.
- **The solution is fractional:** If the objective value is worse than the current incumbent, i.e., lower than the incumbent value for a maximization problem, then the node can be fathomed. Otherwise, a new fractional variable is selected and the process is repeated.

In this way, the Branch & Bound algorithm searches the binary search tree. If fathoming is not used and there are N binary variables, $2^{N+1} - 1$ nodes need to be searched, i.e. LP optimized. The fully searched tree is shown below. Notice that due to fathoming, only 11 nodes are investigated. The full tree would contain 63 nodes. One of these leads to an integer solution with an objective value of 25, the optimal solution. The red nodes

are infeasible, and the remaining non-integer green nodes have optimal values less than 25 and therfore do not need to be explored.



Modern MIP solvers have evolved tremendously over the last 40 years. The description here is only meant to provide intuition behind the procedure.

5.2.1 Making solvable models

How hard a MIP model is, depends on both the size and formulation of the MIP model. More specifically, there are two aspects which are very important for the solvability of the model:

The number of integer/binary variables: The height of the Branch & Bound tree depends directly on the number of integer/binary variables. Two things can be done:

- Some integer variables will take integer values, even if not forced to. Then it should be formulated as a continuous variable.
- Integer variables for which it is possible to deduce their value should be fixed to this value, e.g., $x_{1,5} = 1$. The easiest way to do this in Julia/JuMP is:

```
fix(x[1,5],1; force = true)
```

MIP Gap: The difference between the optimal value of the MIP model compared to the value of the optimal **relaxed** MIP model, where all the integer/binary variables have been converted to continuous variables. The size of this difference is critical. To find this gap, solve the first MIP model and compare to the optimal relaxed model which can relatively easily be found, using the following model:

```
relax_integrality(model)
optimize!(model)
```

If several different models are available for a problem, which often happens, we can compare the models and choose the best model, wich is often the model with the fewest integer/binary variables and/or the lowest gap.

5.3 Solution hardness

The Branch & Bound algorithm for solving MIP models was invented by Ailsa H. Land and Alison G. Doig in 1960 Land and Doig (1960). Today MIP models are one of the most important tools applied in Operations Research. Importantly however, solving them is much harder than solving LP models. LP models can be solved with millions of variables; however, even MIP models with fewer than 100 variables may not be solvable to proven optimality. There are deep theoretical reasons for this difference, which we will briefly discuss in Chapter 10. However, since the invention of the Branch & Bound algorithm, researchers have continued to improve the approach and in particular in the last 30 years, the size of the MIP models which can be solved to optimality have improved tremendously. This has gradually enabled the use of MIP models to solve large real-world problems. A behaviour which is observed for all MIP models is that the solution time will grow **exponentially** with the size of MIP model, i.e. with the number of integer variables in the MIP model. In Figure 5.1 below, a MIP model for different

sized Travelling Salesman Problems (TSPs) is solved to optimality. The model of TSP tested is the model from the Grocery problem 6.15. Five different solvers are considered: HiGHS (light blue), CPLEX (grey), Gurobi (yellow), Cbc (dark blue) and GLPK (red) and they are all tested on 20 different random TSP problems for each size. All solvers show the same behaviour: Smaller models can be solved fast but when the size increase, at a certain **bending point**, the solution time suddenly increases dramatically. This kind of bending point exists for all MIP models and for all solvers, but as can be seen in Figure 5.1, it happens first for the GLPK solver, then the HiGHS solver, then the Cbc solver, then the CPLEX solver and finally the Gurobi solver. However, given a MIP model, if the model is of sufficiently small size, the solvers can handle the problem.

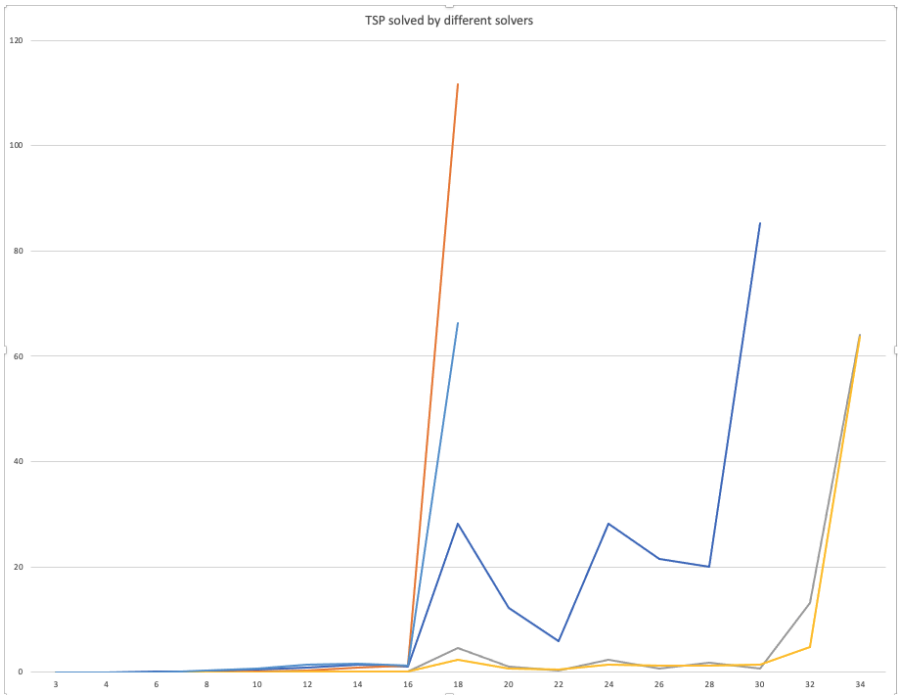


Figure 5.1: Average solution time of TSP with different solvers

Looking at the above graph, you may be wondering why we choose to prefer the HiGHS solver in this book. The answer is that in the future, the HiGHS solver is going to be further developed whereas the development of the Cbc solver seems stalled. Furthermore, the printout from the HiGHS solver is much better.

6 Mixed Integer Programming problems

This chapter contains MIP exercises. All of the assignments require the use of discrete decision variables and thus the formulation of MIP models. The problems considered appear in increasing order of difficulty.

6.1 Young Couple [1]

A young couple, Eve and Steven, want to divide their main household jobs (cleaning, cooking, dish-washing, and laundry) between themselves so that the total time they spend on household duties is kept to a minimum. Their efficiencies on these tasks differ, and the time each needs to perform the tasks is given in Table 6.1:

	Cleaning	Cooking	Dishwashing	Laundry
Eve	4.5	7.8	3.6	2.9
Steven	4.9	7.2	4.3	3.1

Table 6.1: Task times (hours)

Assignment 1.1

Formulate a binary mixed integer model that minimizes the total number of required hours for household jobs and ensures that all jobs are done by someone. Implement and solve it in Julia/JuMP.

Optimal objective value for Young Couple Problem: 18.2 hours

Assignment 1.2

Is the job assignment fair? Make a new program where Eve and Steven perform the same number of tasks.

Optimal objective value for Young Couple Problem: 18.4 hours

[1] Hillier and Lieberman: Introduction to Operations Research"

6.2 Startup Fund

A startup fund manager decides to use MIP models in order to optimize their profit. A new investment fund is just starting up, and €100 million is available for investments. The startup fund staff have identified nine possible investment opportunities in startup companies, the core information being the required capital to make the investment and the expected profit this will give at the end of the investment period.

	Investment								
	1	2	3	4	5	6	7	8	9
Estimated profit	29	35	24	52	53	41	43	68	28
Capital required	17	25	19	25	28	23	29	31	18

Table 6.2: Investment details (millions of dollars)

Given an available budget of €100 million, the objective for the fund is to select the best possible set of investments such that the total estimated profit is maximized. Since these investment opportunities are in startup companies, it is only possible to invest or not invest, i.e. it is not possible to pay a fractional amount of the capital required, nor to pay more than the capital required. Money, which is not invested, is sent back to the investors in the Startup Fund and do not count towards the profit.

Assignment 2.1

Formulate a binary integer program that can be used to maximize the total earned profit and implement it in Julia/JuMP.

Optimal objective value for Investment Optimization Problem: €191 million

Assignment 2.2

After creating the first investment plan, the manager decides that investments in 1 and 5 are competing companies, so it is not possible to invest in both companies. Furthermore, the investments in 6 and 9 can only be performed if either investment 2 or investment 3 is done, since the developed technology in these is the basis of 6 and 9.

Optimal objective value, maximizing the total earned revenue, for Investment Optimization Problem: €185 million

6.3 Logic

This assignment should be done on paper. You have to relate three binary variables x, y and v.

Assignment 3.1: OR

Define (two or three) constraints, which relate the variables x and y to v such that if either x or y is 1 then v is forced to be 1. The truth table for the OR boolean operator is given below:

x	y	v
0	0	0
1	0	1
0	1	1
1	1	1

Assignment 3.2: AND

Define (two or three) constraints such that if both x and y is 1 then v is forced to be 1. The truth table for the AND boolean operator is given below:

x	y	v
0	0	0
1	0	0
0	1	0
1	1	1

Assignment 3.3: NOT

Define one constraint such that if x is 1 then v is forced to be 0 and if x is 0 then v is forced to be 1. The truth table for the NOT boolean operator is given below:

x	v
1	0
0	1

6.4 Stamps

Your uncle Archibald, whom you have not seen for 20 years, dies, and in his will, to your big amazement, lets you inherit his stamp collection. Your only problem is that you have absolutely no interest nor knowledge about stamps. Some of the countries from where the stamps originate, you have not even heard of!

Initially you consider whether to simply scrap the entire stamp collection of 100 stamps. Before you do so, you do, however, make a quick internet picture search for the stamps, and suddenly realize that at least some of the stamps are rather valuable! Therefore you instead decide to make some money, but you have zero knowledge about the stamp market. You consider to contact the local stamp collecting club but realize that your very limited knowledge makes it easy to trick you into a bad deal.

A friend comes up with a brilliant suggestion: Make an online auction! Since at least some of these stamps are rather valuable, it makes sense to do a little work. You and your friend make a simple website, where you put pictures of all of the 100 stamps, numbered from 1 to 100. You then send out emails to a number of stamp-collector clubs and ask for email-bids before a certain date.

A new problem then arises: Stamp collectors typically collect complete sets of stamps, e.g. all Danish stamps from 1977. Thus, if a stamp collector lacks two stamps to complete a collection, then he/she wants to buy *both* stamps or none of them. You receive 213 bids (in €) for different subsets of stamps from your set of 100 stamps. Each bidder wants *all* stamps in their bid at the price they offer. In other words, if they cannot get all of the stamps in the bid, then they will not have any.

The data for the problem can be downloaded from the book web-site `https://www.man.dtu.dk/MathProgrammingWithJulia`. The data, in Julia format, are:

- $BidPrice_b$: The price offered in bid b in € for a set of stamps.
- $BidSets_{b,s}$: Incidence matrix, where $BidSets_{b,s} = 1$, if bid b includes stamp s and 0 otherwise.

You can then either simply copy-paste the data into your program or you can use the `include(<file name>)` command.

Assignment 4.1

Formulate the problem of choosing which bids to accommodate, such that your profit is maximized, as an integer program and solve it using Julia/JuMP.

Optimal objective value for the stamp problem: €11084

You are very happy about the earnings but, to your surprise, not all stamps are sold! You discuss this with a friend, and he considers this a stupid loss, and blames the outcome on poor optimization. Hence, he suggests to *require* that all stamps are sold.

Assignment 4.2

Make the new model, what happens? Why? And is the suggestion good?

6.5 Micro brewery 2

Encouraged by your good results as a beer producer, you realize that if you offer a broader selection of beers, you may be able to sell more beers. So after some trial production, you decide to start the production of three types of beer: TSP-Stout, Knapsack-Dark, and Set-Partitioning-Light. After some tastings, the cafes order different amounts of beer. The total amounts ordered for each month and of each type are shown in Table 6.3:

	TSP-Stout	Knapsack-Dark	Set-Partitioning-Light
jan	35	15	5
feb	20	10	20
mar	15	20	20
apr	45	15	35
may	25	15	35
jun	65	55	80
jul	40	90	60
aug	50	80	30
sep	35	25	35
oct	85	45	20
nov	50	5	20
dec	55	30	40

Table 6.3: Beer demand (litres)

You have the same brewing system, with a capacity of 120 litres, but you can only brew **one** type of beer per month! Since your mother wants to encourage your business, she allows you to store 300 litres of the beer types combined (i.e., the sum of stored beer over all tree types can at most be 300 per month). She also reduces the storage cost to €0.1 per liter stored per month, for all beer types. You have produced an initial storage with 25 litres of TSP-Stout, 65 litres of Knapsack-Dark, and 75 litres of Set-Partitioning-Light.

Assignment 5.1

Make a Mixed Integer Linear Programming model, which plans the production and storage for the whole year to meet demand optimally, minimizing the storage costs, and taking into account that only one type of beer can be produced per month. Implement the Microbrewery model in Julia/JuMP.

Optimal objective value for Micro brewery 2 Problem: 192.5

6.6 Santa's Workshop Tour 2019 [2]

♪ ♪
Hammers ring, are you listenin'
In the shop, toys are glistenin'
Should they see the sights?
There might be a fight
Walkin' 'round the Workshop Wonderland

Families said, they want to see it
Santa said, he'd guarantee it
They pick a date
But they may have to wait
Walkin' 'round the Workshop Wonderland

We told Santa that he was a madman
He just wants to make sure they all smile
He'll say "Are you flexible?", They'll say "Yeah man,
But can you help us make it worth our while?"

"Give them food, or sweater
the more they wait, the gifts get better"
Please help us rank
Or we'll break the bank!
Walkin' 'round the Workshop Wonderland
♪ ♪

Santa has exciting news! For 100 days before Christmas, he is opening up tour of his workshop. Because demand is so strong, and because Santa wanted to make things as fair as possible, he let each of the 5,000 families that will visit the workshop choose a list of dates they'd like to attend the workshop.

Now that all the families have sent Santa their preferences, he's realized it's impossible for everyone to get their top picks, so he's decided to provide extra perks for families that don't get their preferences. In addition, Santa's accounting department has told him that, depending on how families are scheduled, there may be some unexpected and hefty costs incurred.

Santa needs the help of the 42112 community to optimize which day each family should

[2] This comes from the traditional Christmas optimization challenge, see: https://www.kaggle.com/c/santa-workshop-tour-2019

get to go to the workshop in order to minimize any extra expenses that would cut into next years toy budget! Can you help Santa out?

Your job is to select the visit dates for each of the 5000 families. Each family consists of a number of persons and each family *HAS* to get a visit. Each family has given a list of wishes regarding visit dates, but these are NOT included in the data! Instead, these have been used to calculate DayVisitCosts for Santa Claus. This calculation is rather complex, but the calculations have been performed for you. See the details on the Kaggle website if your are interested. Each day there has to be at least 125 visitors and there can be at most 300 visitors.

When solving this problem, the difference between an open-source solver like HiGHS and a closed-source solver, e.g. Gurobi will be prominent. The free solvers cannot solve the full problem with 5000 families and 100 possible visit days days within a reasonable time. For this reason there are two data-sets for this problem: One with 5000 families and 100 days and a reduced data-set with 1000 families and 20 days. The data-set files are:

- SantasWorkshopData_5000_100.jl: The full data-set, with 5000 families and 100 days.
- SantasWorkshopData_1000_20.jl: The full data-set, with 1000 families and 20 days.

The data-set files defines 4 julia variables:

- F : The number of families
- D : The number of days for the visits (100 or 20)
- $FamilySize_f$: The number of members of family f
- $DayVisitCost_{f,d}$: Cost of assigning family f to day d. These values include the cost of compensations for families who do not get assigned to a preferred day.

Also, the objective tolerance of the Gurobi solver may actually lead to a solution which is not precisely optimal, because the default objective tolerance is 0.01 % . This can be handled by setting the MipGap parameter for the solver, see Section 3.7

Assignment 6.1

Formulate a mathematical model that can be used to optimally schedule the family visits at minimal cost. Implement your model in Julia/JuMP.

Optimal objective value for Santa's Workshop Tour 2019, 1000 families over 20 days: 164068

Optimal objective value for Santa's Workshop Tour 2019, 5000 families over 100 days: 43622

6.7 Factory Planning 2 [4]

Instead of stipulating when each machine is down for maintenance in the factory planning problem, it is desired to find the best month for each machine to be down. Refer to Section 4.5 all previous information.

Except for the grinding machines, then all machines must be down for one month during the six month period. So for example one vertical drill could be down in January and the other in April or they could both be down at the same time in March etc. For the grinding machines it is only necessary for two of the four machines to be down during the whole six month period.

Assignment 7.1

Extend the previous model so it is up to the program to decide when the maintenance of each machine should take place rather than being pre-decided. You should still maximize profit as before. What is the value of the extra flexibility of allowing machine downtime to be chosen?

Optimal objective value for Factory Planning Problem: £108855 (ultimo storage)

Optimal objective value for Factory Planning Problem: £109030 (primo storage)

[4] Taken from Williams: "Model Building in Mathematical Programming"

6.8 Student Teacher Meetings

In Danish schools, it is customary to have meetings during the school year, often twice a year, between the student and parents and some of the teachers of the student. The meetings take place in the evening. Before the student-teacher meeting evening, the student (or parents!) make a list of the teachers they would like to meet with. When all students have submitted their teacher meeting request, a schedule needs to be made.

The teachers each use their own classroom for their meetings. The students and their parents walk around the school and meet with the teachers they wish to talk to. A number of time slots are given, e.g., a 10 minute interval, and your job is simply to plan when, i.e. in which time slot, the meetings will occur.

You can find data for the assignment on the book web-site `https://www.man.dtu.dk/MathProgrammingWithJulia`. Six different problem instances are available, where StudentTeacherMeetingData_15_6.jl have 15 students and 6 teachers:

```
StudentTeacherMeetingData_15_6.jl
StudentTeacherMeetingData_20_6.jl
StudentTeacherMeetingData_25_10.jl
StudentTeacherMeetingData_25_8.jl
StudentTeacherMeetingData_30_10.jl
StudentTeacherMeetingData_40_10.jl
```

In each file, the following information is given:

- $StudentTeacherMeetings_{s,t}$: An incidence matrix, with one row for each student s and one column for each teacher t. If student s should meet teacher t, then $StudentTeacherMeetings_{s,t} = 1$, otherwise $StudentTeacherMeetings_{s,t} = 0$
- S: Number of students
- T: Number of teachers
- TS: Number of time slots

The meetings are private, so a teacher can only meet one family at a time. The families would like to spend as little time as necessary on the meetings. Therefore, the overall objective is to minimize the sum of the latest time slot indices assigned to the families.

The problem is surprisingly difficult, so the data sets are all rather small and open-source solvers all struggle on even the smallest instances. This problem thus shows the value

of the expensive commercial solvers CPLEX and Gurobi.

As described Section 3.7, it is possible to set a maximal solve time using

```
set_time_limit_sec(model, 600) # timelimit of 600 sec (5 minutes).
```

Note that the solver will not stop immediately after 600 seconds sharp, but wait to finish the current iteration of Simplex or what else it is doing. With a time limit on the solver you will likely not manage to find the optimal solution, but you can check the value of the best solution that has been found (if any) using the usual `objective_value(model)`. You can then compare this with the dual bound (lower bound for minimization) at the time the solver stopped: `objective_bound(model)`. The optimal solution will have a value somewhere between these two.

Assignment 8.1

Formulate the student-teacher meeting planning problem as a MIP, minimizing the sum of late time slots. Implement your model in your Julia/JuMP.

Only the optimal objective values for the first five problems are reported (together with the Gurobi solution time). These results were obtained using a Mac Book Pro, 2018.

Optimal objective values for the student-teacher meeting problems

- StudentTeacherMeetingData_15_6.jl : 66 (sec. 52.9)
- StudentTeacherMeetingData_20_6.jl : 110 (sec. 632.6)
- StudentTeacherMeetingData_25_8.jl : 131 (sec. 1393.3)
- StudentTeacherMeetingData_25_10.jl : 113 (sec. 1345.4)
- StudentTeacherMeetingData_30_10.jl : 152 (sec. 603.7)
- StudentTeacherMeetingData_40_10.jl : 248 ($\geq$ 244, 1.61% gap) (sec. 929.7)

Here

6.9 Workplan for Teaching Assistants

A Mathematical Programming crash course, over 9 days, is taught by 4 teaching assistants (TAs) and 2 professors. The professors give the lectures and the TAs are available for help with the exercises. Suppose that four TAs (AL, FR, JE, and MI) are the TAs in the course. In this assignment, you have to plan when they should work during the course. Each TA must work exactly 52 hours (a total of 208 hours are available for the four TAs). The professors have attempted to come up with a plan of how many TAs to assign for exercise help in each of the 8 time slots from 9 to 17. This TA demand is given below in Table 6.4.

	Day								
	1	2	3	4	5	6	7	8	9
9-10	4	4	4	0	3	0	3	0	2
10-11	4	4	4	2	4	2	4	2	2
11-12	4	4	4	4	4	4	4	4	2
12-13	4	3	3	3	3	4	3	3	1
13-14	4	3	3	3	3	4	3	3	1
14-15	4	3	3	3	3	4	3	3	1
15-16	3	3	2	3	3	4	3	3	1
16-17	3	3	2	2	2	3	2	2	1

Table 6.4: TA demand for the first 9 days

To make the work easier for the TAs, we wish to take into account their personal preferences. Each TA has specified inconvenience scores for each day d and time period t in the matrix of inconvenience scores $Inconvenience^{ta}_{t,d}$. You can find these data on the book web-site `https://www.man.dtu.dk/MathProgrammingWithJulia`. You can then either simply copy-paste the data into your program or you can use the `include(<file name>)` command. Notice that the data also contains a simple normalization of the Inconvenience matrixes.

Assignment 9.1

Formulate a mathematical model that can be used to find the optimal work plan for the TAs. Implement your model using Julia/JuMP to find the optimal solution.

Optimal solution TA work plan (minimal inconvenience): 94.97

When you inspect the generated plan, you observe that the TAs work several times on each day. This is rather inconvenient. You decide to modify your model and include the restrictions that each TA can only work consecutive hours and that if a TA works on a particular day, then the TA has to work at least two hours.

Assignment 9.2

Revise the mathematical model with additional restrictions. Implement this model using Julia/JuMP to obtain the optimal plan.

Optimal solution TA workplan (minimal inconvenience): 121.97

Notice that while these extra requirements are very reasonable, it leads to a worse solution, regarding the TA in-convenience. On the other hand, the revised solution is an improvement for the TAs in a different way.

6.10 Scrap Removal

After cleaning up in the shed (Project scheduling), your father has identified a number of scrap items, which he needs to get rid of. There are 20 items of different weight. Your father contracts a haulage contractor, to come and pick up big-bags of scrap. The contractor is paid €50 for each big-bag he has to transport away. Each big-bag can carry at most 100 kg of scrap and your father has ordered 10 big-bags. He makes the following deal with you: If you pack the items in big-bags, not breaking the 100 kg limit, and are able to use fewer big-bags, then **you** will get the €50 for each big-bag saved. How much money can you make?

	Items									
	1	2	3	4	5	6	7	8	9	10
weight (kg)	35	10	45	53	37	22	26	38	63	17

	Items									
	11	12	13	14	15	16	17	18	19	20
weight (kg)	44	54	62	42	39	51	24	52	46	29

Table 6.5: Weight for each of the 20 items

Assignment 10.1

Implement a MIP model in Julia/JuMP which minimizes payment for the contractor.

Optimal objective for Scrap Removal: €400

6.11 Rolling Stock Scheduling

This exercise considers the rolling stock scheduling problem as described by Alexander Schrijver, *Minimum Circulation of Railway Stock*, 1993 (for short "Schrijver" in the remaining). In Schrijver the number of passengers per trip is fixed.

You will implement Schrijver's model with two rolling stock types in Julia.

For your convenience, you can download *RollingStock-Data.jl* from DTU Learn. This file contains a data array "Arcs", where the rows represent arcs in the model of Schrijver, and the columns represent respectively:

- tail node of arc (from node)
- head node of arc (to node)
- indicator: 1 if the arc is part of A_0 (overnight arc), 2 if the arc is part of A^r (a traintrip arc), and 0 otherwise (e.g. in station flow arcs that are not representing the overnight flow)
- 1 columns with first class demand
- 1 columns with second class demand

You may use *RollingStock-Data.jl* as starting point for your Julia implementation. In addition, there is the file Nodes.txt that contains a description per node. If you would like to link the name of a station to a node, than you can use this file.

In the following questions, you can note the following: the mathematical model formulation, the solution (in terms of the number of typeIII and typeIV units, other information where relevant), and the solution times (of Cplex).

1. Formulate the rolling stock problem of Schrijver.
2. Implement the above model, and check that you obtain the same solution.
3. Change the model such that you minimize the number of stations used for parking overnight, and solve it. How high should the penalty be to use one less station than in the unconstrained model? You can find an answer through trial-and-error.
4. Change the model such that there is a start up cost for using a station for overnight storage. Furthermore, there is flexible cost for crew member. Per every 5 train units, you need 1 person crew member. That is, for 1-5 trains you need 1 crew member, but for 6-10 train units stored you need 2 crew members, and so on. Formulate the new model, and implement it. Motivate your choice for penalty costs of the crew members.

5. Change the model such that you allow to not serve all second class demand. A seat too little will cost 100 euros. Does this change the solution? Try to find the minimum penalty for which allowing seat shortages changes the optimal solution.

In the previous, we have created one rolling stock schedule. However, often demand is different per day. Let's change our problem to having 5 different demand scenarios that you can find in "RSdataMultipleDays.jl". For this exercise, let's assume we can use a composition length of maximum 17 (rather than 15 before).

6. Formulate a model that minimizes RS purchase costs and driven kms per unit for the 5-day schedule. The solution may assign different rolling stock units to arcs on different days, but the overnight-stays should always be the same: e.g. the number of units of a specific type that stay overnight at a specific station has to be the same over all days.

7. Implement this model. Motivate your choice for the cost of driven kms per unit. For ease of implementation, you may assume that every traintrip arc is equally long, while station flow arcs and overnight arcs of course have length 0.

6.12 Tariff Rates [6]

A number of power stations are committed to meeting the electricity load demands given in Table 6.6. The demand given refers to the whole time slot. As an example, 15000 MW of power are needed in the first time slot.

Time slot	Demand (MW)
0.00 - 6.00	15000
6.00 - 9.00	30000
9.00 -15.00	25000
15.00-18.00	40000
18.00-24.00	27000

Table 6.6: Time slot demand

There are 3 different types of electricity generators available. There are twelve of type 1, ten of type 2, and five of type 3. Each generator has to work between a minimum and a maximum level. There is an hourly cost of running each generator at the minimum level. In addition, there is an extra hourly cost for each megawatt at which a unit is operated above the minimum level. To start up a generator also incurs a fixed cost. All of this information is summarized in Table 6.7.

In addition to meeting the estimated load demands, there must be sufficient generators working at any time to make it possible to meet an increase in the load of up to 15%. This increase would have to be accomplished by adjusting the output of generators already operating within their permitted limits.

	Minimum level	Maximum level	Cost/hour at minimum	Cost/(hour * MW above minimum)	Start-up cost
Type 1	850	2000	1000	2	2000
Type 2	1250	1750	2600	1.3	1000
Type 3	1500	4000	3000	3	500

Table 6.7: Generator type specifications

[6] Taken from Williams: "Model Building in Mathematical Programming"

Assignment 12.1

Formulate a mathematical model to and use Julia/JuMP to find the best daily electricity production plan. This plan will be used day after day (i.e. a *cyclic* production pattern). Which generators should be working in which periods of the day to minimize total cost?

Optimal solution for Tarif Rates Problem: 988540

Assignment 12.2

What would be the savings of lowering the 15% reserve output guarantee? In other words, what does this security of supply guarantee cost?

Optimal solution for Tarif Rates Problem: 988540-987790=750

Tip 1: Notice that you do not need to know *which* specific powerplant is running, only *how many* of the different types are running.

Tip 2: If you get a solution with optimal value 1015150, think about that the production pattern is *cyclic*.

6.13 Railway Depot Shunting

Passenger trains usually consist of multiple *units* coupled together. Units are categorized by unit types, where units of the same type have the same physical attributes (length, seat capacity, etc.). Over the course of a given planning horizon train units are coupled and decoupled from trains at certain depots to match the capacity of the trains in service to the expected passenger demand. Units not in service are parked (or stored) at depots, where they await their next service. This idle time at the depot can be used for maintenance purposes, among other things. Depots typically comprise multiple parallel tracks (which are not necessarily the same length) that can often only be accessed from one direction. In other words, the tracks operate as Last-In-First-Out (LIFO) stacks. For a given time horizon, checking whether all units that need to be parked can feasibly do so determines whether the proposed train services are ultimately feasible.

You have been asked to consider one depot that contains eight LIFO tracks and investigate whether or not 70 units that have been assigned to the depot can park there overnight. There are two types of units: Type One is 42m long, while Type Two is 84m. Each unit has a known arrival time in minutes after midnight(when it is taken out of service) and a known depot departure time (when it is needed back in service), again in minutes (but adjusted for midnight). These types have been made available in the data file `unit_arrival_departure_info.txt`. The respective track lengths are specified in Table 6.8.

	Track							
	1	2	3	4	5	6	7	8
Length (m)	500	800	700	400	650	650	750	500

Table 6.8: Track Lengths

Your aim is determine whether it is possible for all units to feasibly park in the depot. On which track will you park the different units? You may assume that a unit occupies depot capacity at its arrival time but has released depot capacity at its departure time. Two units are in conflict if one blocks the departure of another. Note that it is usually not possible to swap the departures of two units of the same type that are in conflict since each has unique maintenance requirements that must be adhered to.

Assignment 13.1

Formulate a mathematical model that can be used to determine the maximum number of units that can be feasibly parked in the depot. Implement your model in Julia/JuMP to determine what this value is.

Optimal objective value: 64

Assignment 13.2

You have now been asked to investigate the impact of ensuring that only units of the same type can park on the same track. Modify your formulation to ensure this. What is the maximum number of units that can now be parked? Implement your model in Julia/JuMP to determine what this value is.

Optimal objective value: 53

6.14 Tennis

If you play tennis on a clay tennis court, then you must prepare it for the next couple of players. This is done by first brushing the clay and then sweeping the lines, using a special broom, see Figure 6.1.

Figure 6.1: Sweeping a tennis line

This assignment deals with the problem of finding the easiest (shortest) way of sweeping the lines (of one side) on the tennis court. In Figure 6.2, the outline of a tennis court is given, with all measurements in feet.

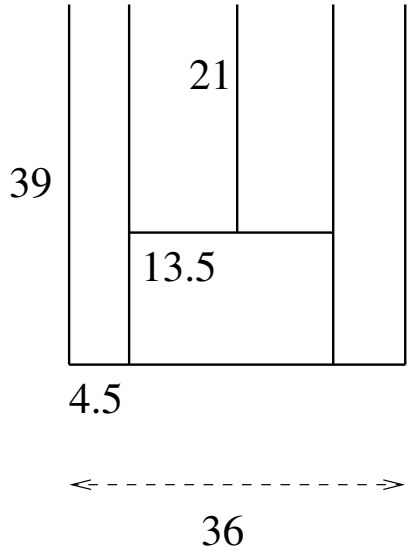


Figure 6.2: Court measurements (feet)

The objective is to minimize the walking distance when sweeping the lines. When solving this problem, different formulations are possible.

Hint: Define a coordinate system and let all of the places where lines start or connect to other lines be points in the coordinate system. There are 12 such points. Then, calculate a distance matrix, which can easily be calculated in a couple of for loops.

Assignment 14.1

If you are *only* allowed to step on the lines, then what is the best route you can walk, if you have to return to the starting position? This can be calculated analytically.

Optimal distance: 390 feet

Assignment 14.2

Construct a mathematical model to find the best route if you are allowed to walk between the lines, but still have to start and stop at the same points. Implement your model in Julia/JuMP.

Optimal distance: 306 feet

Assignment 14.3

Relax the model such that it is allowed to start and stop at different places. What is the best route now?

Optimal distance: 280.5 feet

Assignment 14.4

Constrain the model such that the start and stop places are only allowed on the outside lines (the 3 outside lines being the leftmost line, the bottommost line and the rightmost line in Figure 6.2. What is the best route?

Optimal distance: 287.47 feet.

6.15 Groceries

The local grocer in your neighbourhood decides to offer delivery of goods to home and offers you the job of driving around with the deliveries. The job is simple: You go to the grocery shop 16.00, get the list of persons to deliver to and the bags of groceries in. You then make a round-trip in the grocery van, visiting all the customers, delivering their goods. When you return the van to the grocery shop, your work is over. You are paid a fixed amount each day, so the faster you can deliver the goods, the faster your job is over.

Since you want to work as little as possible, you want to plan the shortest possible route. Given 5 customers, with the coordinates below in Table 6.10, where the grocery is in origo (0,0). We will for simplicity assume that direct travel between the different points is possible, and simply use euclidean distance. How can you formulate a Mixed Integer Programming model to minimize the total round-trip tour length ?

Customer	X coordinate	Y coordinate
Grocery	0	0
1	104	19
2	370	305
3	651	221
4	112	121
5	134	515

Table 6.9: Grocery and customer coordinates

You quickly realize that any round trip will consist of a number of different decisions, i.e. whether to travel from one customer to another or not. Each such decision can be defined using a binary decision variable $x_{i,j} \in \{0, 1\}$, which is 1 if you travel from customer i to customer j. This makes it simple to calculate the total travelled distance: $\sum_i \sum_j Dist_{i,j} \cdot x_{i,j}$. This is obviously the objective function.

However, how can you define constraints so that only solution values of $x_{i,j}$ which represents correct round trips are feasible ? First you realize that any round-trip tour, means that you travel exactly once from one of the other customers j **to** a customer i and **from** a customer i to one of the other customers j. These two requirements can be implemented with two abstract constraints. It is also necessary to fix the variables $x_{i,i}$ to zero, i.e. going from city i to city i is not really interesting, so the value of these variables needs to be fixed to zero.

Assignment 15.1

Formulate the first model with the objective function to be minimized and the two different abstract constraints, ensuring that you travel to each of the customers and leave each of the customers.

Optimal objective value for the grocery problem: €1576.85

When you print out the values of the binary $x_{i,j}$ variables, you do, however quickly realize that this is not a feasible solution, since it consists of two separate round-trips. But the solution **is** legal according to your model. Therefore, you need one or more additional constraints, which will make separate round-trips impossible. How to do that?

One of your friends suggests the following: Assign a counter (a decision variable) $u_i >= 0$ to each customer. This can simply be a positive continuous variable. Then create a constraint, which ensures that whenever you go from one customer to the next, the value of the counter is at least one unit larger than the customer you come from. At first this seems easy: $x_{i,j} + u_i \leq u_j \; \forall i, j$. This constraint does really do the job, when $x_{i,j} = 1$, but it also introduces a lot of relations between customer i and customer j when $x_{i,j} = 0$, and that is wrong. So the constraint needs to be altered such that it does not have any influence on the solution, unless $x_{i,j} = 1$. Therefore, you make a little trick. Let C be the number of customers (and the Grocer) in the model. Then re-write the constraint to: $u_i + 1 \leq u_j + C \cdot (1 - x_{i,j}) \; \forall i, j$. Notice, that if $x_{i,j} = 1$, we get exactly the previous formula, but if $x_{i,j} = 0$ we add a large number on the left hand side, to allow any value of u_j.

However, if you simply add this constraint, no solutions will exist, because for each tour, **one** starting point is necessary, to go from a high count to 0. But, this is exactly the point: If you only allow not to update the counter for one customer, e.g., the grocery, only one tour will exist. Hence, the constraint should hold for all pairs of customers, **except** one, e.g., the first one.

Assignment 15.2

Formulate the second model where you extend the previous model with counting variables and constraints which forces the counter variable to be one unit higher than the previous customer on the tour.

Optimal objective value for the grocery problem: €1857.51

After a one month trial period, the grocer opens up for taking more customers, and it is a great success, now there are suddenly 13 customers, coordinates given below, in

combination with the size of the delivery size. The capacity of the van is 26.

Customer	X coordinate	Y coordinate	grocery size
1	0	0	0
2	104	19	3
3	370	305	9
4	651	221	7
5	112	121	11
6	134	515	11
7	797	424	6
8	347	444	7
9	756	141	7
10	304	351	2
11	236	775	4
12	687	310	2
13	452	57	8

Table 6.10: Grocery and customer, coordinates, and delivery size

This, however presents a new problem: There is simply not room in the small van for all the groceries. Actually 3 vans of this size are required. So the grocer decides to acquire 2 more vans of the same size and hire 2 more delivery workers. But now you have a problem: Your nice model only works with **one** van.

The model needs to be updated to accommodate 3 vans. Since all the vans start going out of the grocery parking lot, the sum over the $x_{1,j}$ variables, from the grocery the customers should be equal to 3, and 3 should be going into the grocery node $x_{j,1}$. Instead of the counting variable counting units, we now count how much material has been delivered at this point of the route, i.e. $u_i + q_i \leq u_j + Q \cdot (1 - x_{i,j})$, where Q is the capacity of the van, here 26. this problem may take significant time using the open-source solver HiGHS.

Assignment 15.3

Formulate the third model where you extend the previous model to handle 3 vans.

Optimal objective value for the grocery problem: €5063.35

6.16 Aircraft Landing

Aircraft movements at large airports are subject to a number of security constraints. You have been asked to schedule the landing sequence of ten aircraft at a busy European hub. Each aircraft has an earliest arrival time (corresponding to flying at its maximum speed) and a latest arrival time (mainly influenced by its fuel supplies). Within this window the aircraft has a target time, or the ideal arrival time of the aircraft. Due to the disruption it causes, any deviation from this target time (forwards or backwards) incurs a penalty per minute. Table 6.11 contains all relevant aircraft data.

Aircraft	1	2	3	4	5	6	7	8	9	10
Earliest Arrival	125	187	94	99	111	123	129	131	142	155
Target Time	155	212	134	127	123	138	145	157	162	179
Latest Arrival	473	612	508	421	511	515	525	407	500	612
Earliness Penalty	10	20	35	25	35	30	30	50	30	35
Lateness Penalty	10	20	35	25	35	30	30	50	30	35

Table 6.11: Aircraft data

Aircraft	1	2	3	4	5	6	7	8	9	10
1	-	5	15	15	15	15	15	10	15	5
2	5	-	15	15	10	15	10	15	15	15
3	15	15	-	10	15	10	15	10	15	10
4	15	15	10	-	10	10	10	15	15	15
5	15	10	15	10	-	10	5	10	15	5
6	15	15	10	10	10	-	15	15	15	15
7	15	10	15	10	5	15	-	10	5	10
8	10	15	10	15	10	15	10	-	10	15
9	15	15	15	15	15	15	5	10	-	10
10	5	15	10	15	5	15	10	15	10	-

Table 6.12: Required aircraft separation time (minutes)

Due to turbulence and the time needed for an aircraft to clear the runway, a safety interval (or separation time) is needed between any two landings. Table 6.12 indicates the required separation times between two specific aircraft. This depends on, among other things, the sizes of the two aircraft. An entry in line p of column q states the minimum time needed between the landing of aircraft p and the subsequent landing of aircraft q, even if this not consecutive.

Assignment 16.1

Formulate a mixed integer linear program that can be used to find the landing sequence that minimizes the total penalty incurred. Assume that landing times are required to be integer. Implement the model using Julia/JuMP to find an optimal solution.

Optimal objective value: 2025

Assignment 16.2

You have been asked to modify your solution to ensure that aircraft 5 and 6 do not land consecutively in either order. Modify your formulation above to ensure this and, using Julia/JuMP, find the best landing sequence satisfying the additional constraints.

Optimal objective value: 2080

Assignment 16.3

Consider the problem given in 17.2. You have now been asked to find the second best landing sequence. Modify your formulation to ensure that the second best landing sequence is obtained. Implement the model in Julia/JuMP and state the landing sequence (note that the second best sequence may have the same objective value as the optimal one)

Optimal objective value: 2080

6.17 Food Festival

A close friend of your mother is in charge of a food festival in your town, i.e. a number of stands in tents around in the city, where food can be presented and tasted by the festival participants. There is however a planning problem at the festival: During the late evening, night and early morning, security guards are needed to watch the different facilities around the city. There is a list of 25 work shifts which needs to be assigned a security guard. However, some of the shifts cannot be covered by the same security guard because of overlap in time. Below in Table 6.13 is a list of the 25 shifts and for each shift the list of shifts which cannot be done by the same person. If a person is working shift 1 that person cannot work shift 2, shift 4, shift 6, shift 11, shift 17, shift 21, shift 22, and shift 25. Notice, that it goes both ways so a person working shift 25 cannot also work shift 1.

The concern is how many persons it is necessary to hire for the shifts. It is easy to hire 25 persons and then let every person work exactly one shift. This will then not lead to any conflicts. Even though they will all receive the same hourly wage, there are a number of expenses associated with hiring the persons, therefore she wants to hire as few persons for the shifts as possible.

You can find these data on the book web-site
`https://www.man.dtu.dk/MathProgrammingWithJulia`.The data contains:

- $s \in Shifts = \{1, 2, 3, ..., 23, 24, 25\}$
- $Conflict_{s1,s2}$: An incidence matrix such that if $Conflict_{s1,s2} = 1$ then shift $s1$ and shift $s2$ conflicts and cannot be worked by the same security worker.

Assignment 17.1

Formulate a model which minimizes the number of persons it is needed to hire.

Optimal objective value for the Food Festival problem: 5

Shift	Conflicting shifts									
1	2	4	6	11	17	21	22	25		
2	3	7	9	14	16	17	23			
3	4	9	10	11	18	19	22	23		
4	8	9	10	19	20	21	24	25		
5	6	10	11	13	15	16	21	22		
6	8	11	12	13	14	15	16	17	18	19
7	9	10	14	15	21	22	25			
8	9	10	11	15	16	24				
9	12	13	15	19	21	24				
10	11	14	15	16	19	22				
11	12	13	15	19	20	21	25			
12	14	15	16	17	18	23				
13	14	16	19	23	25					
14	15	17	18	22						
15	16	17	23	24	25					
16	18	19	20	21	24					
17	19	22	24							
18	20	21								
19	23	24	25							
20	22	25								
21	22	23	24							
22	24									
23	24	25								
24	25									

Table 6.13: List of shifts and conflicting shifts

6.18 The Wedding Planner

This assignment is about seating guests at tables, during a dinner. With data from a real wedding, G guests are to be seated at T tables. We want to find the best feasible seating-plan such that the guests at the same table share interests, have a somewhat equal distribution of men and women, and that each guest at least know some of the other guests at the table. We will assume that the tables are small, 9 seats, such that everybody at the table can talk to everybody. The feasibility requirements are:

- Each guest should sit at a table
- There is a limit of 9 people at each table
- Couples that attend the wedding should sit at the same table

You can find these data on the book web-site `https://www.man.dtu.dk/MathProgrammingWithJulia`. You can then either simply copy-paste the data into your program or you can use the `include(<file name>)` command. The data contains the following parameters:

- Guests: The set of guests
- G: The number of guests
- Tables: The set of tables
- T: The number of tables
- SharedInterests: A two-dimensional matrix where SharedInterest[g1,g2] specify how many shared interests guest g1 and g2 has.
- Couple: An incidence matrix where Couple[g1,g2]=1 if g1 and g2 are a couple
- Male: Male[g]=1 if guest g is male
- Female: Female[g]=1 if guest g is female
- Know: Know[g1,g2]=1 if guest g1 and guest g2 know each other
- TableCap: No of guests which can maximally be seated at the table, equal to 1.

The final model is rather complex, so we will iteratively build the model in four stages.

Assignment 18.1

Formulate a mathematical model (without an objective) that can be used to solve this problem, i.e. just the constraints, no objective function needed.

Assignment 18.2

Add the following objective function to your model:

Maximize the number of shared interests for persons sitting at the same table

Optimal objective value: 67

Hint: Define a new variable that takes the value of one if two guests are seated at the same table. If you get a higher result, then check to see if you are counting people twice or counting self-shared interests.

Assignment 18.3

Now consider a different objective function. For each table, penalize the number of males and the number of females that are in excess of two. As an example, a table with seven males and two females has an excess of five males and incurs a penalty of three. Minimize the sum of penalties for all tables.

Optimal objective value for: 0

Hint: First formulate two sets of constraints that strictly enforce the requirements that there can be at most 2 too many of each gender. You can then introduce slack variables that allows these restrictions to be broken, but is penalized in the objective.

Assignment 18.4

Now consider a different objective function. You are to minimize the number of unfamiliar connections. The parameter *Know*, given in the datafile, defines which other guests each guests knows. The idea is that each person at the wedding should ideally sit at a table where that person knows at least 3 other people and if they do not know 3 at the table, the number less than three is the penalty.

Optimal objective value: 2

Hint: Make a hard constraint that forces everybody to know at least 3 persons that they sit together with. You can then relax this constraint with a slack variable, which is minimized in the objective function.

When the bride & groom consider the seating plan, they really consider all three objectives *at the same time*. Hence, we have here a *multi-objective optimization problem*. To solve the combined problem we need to evaluate how important each of the objectives is. We consider them to be equally important.

Assignment 18.5

Solve your model again, but this time minimize the sum of the negative number of interests, the number of non-equal male-female tables and the number of unfamiliar connections.

Optimal solution for Wedding Planner Problem: -60

The Wedding Planner problem is actually very very hard. The actual wedding where these data originates had 74 guests. But the MIP models cannot solve the full problem.

6.19 Wedding Planner with Math-Heuristics

This assignment is a training exercise for use of the two math-heuristics: Hamming-distance-heuristic and Fix-and-Optimize heuristic. For this assignmnet, you can download the working code for both algorithms for the wedding planning problem, and you can download the large dataset WeddingData74.jl (the actual wedding plan, with 74 guests). In order to make a good example, there are two changes to the wedding planning problem:

- The objective is to maximize the shared interests. This simplifies the model and problem significantly
- We have added a slack variable which makes it feasible, even if no guests have seats. But it is heavily penalized.

Assignment 19.1

Optimize (maximize) the Wedding Planner shared interest problem with the Hammingdistance heuristic. Which K neighborhood size achieves the best result in two minutes (120 sec.) ?

Assignment 19.2

Optimize (maximize) the Wedding Planner shared interest problem with the Fix-and-Optimize heuristic. Which K neighborhood size achieves the best result in two minutes (120 sec.) ?

6.20 Math-heuristics for Stamps

It turns out that you uncle Archibald did not just leave you 100 stamps, but 400 and the auction got 2000 bids. Solving this with HiGHS or Gurobi is not possible (feel free to try).
In this exercise you are to implement two math-heuristics, Hamming-distance heuristics and Fix-and-optimize heuristics. You can use the math-heuristics from the Heuristic Wedding Planner problem. Replace the model and try to get good solutions. The larger dataset (bid_data_400_2000.jl)is available on DTU-learn.
You can then either simply copy-paste the data into your program or you can use the `include(<file name>)` command.

Assignment 20.1

Optimize (maximize) the Stamp problem with the Hammingdistance heuristic. Which K neighborhood size achieves the best result in two minutes (120 sec.) ?

Assignment 20.2

Optimize (maximize) the Stamp problem with the Fix-and-Optimize heuristic. Which K neighborhood size achieves the best result in two minutes (120 sec.) ?

6.21 Hot Air

This assignment is designed to give you an appreciation of the complexity of a very important real-life problem involving integer variables. It focuses on a simplified version of an airline fleet assignment problem.

Hot Air is an airline company that is considering an aggressive move into new territory in Texas. They have forecasts for the demand, i.e. the number of passengers that wish to travel on Hot Air, for one city to another city. Hot Air now needs to decide which routes to fly, that is, assign aircraft to. Their objective, not unexpectedly, is to maximize profit. Profit is generated by transporting fare-paying passengers, while taking into account any costs incurred when operating aircraft. All the necessary data are given below.

To make the problem manageable, Hot Air has made a number of simplifying assumptions, some more important than others. These are listed below.

Simplifying assumptions:

1. No penalty is incurred for **not** transporting passengers that wish to travel; these passengers will just find another means of travel.

2. All passengers transported must be flown non-stop from their origin to their destination. No change of planes is allowed, nor is staying on a plane that travels via another city (*1-stop routes*). This assumption is relaxed in the last question.

3. Hot Air is seeking a daily schedule. They operate (like most airlines) the same schedule every day. A day is assumed to consist of 3 time periods, where planes can fly from any city to any other city: Dawn to mid-morning, mid-morning to afternoon, and afternoon to evening.

4. Due to restrictions on hangar lease and maintenance/service personnel (who work during the night), a specific number of planes must spend the night at each city. That number, of course, is the number of planes that depart each morning and return each evening.

5. Planes do not have to fly. A plane not flying does not incur any cost.

6. Planes are fungible (interchangeable): Hot Air essentially flies only one type of aircraft. Therefore, only the *number* of planes at any position at any time is important, not the identity of the planes.

7. More than one plane can fly a given route at the same time.

The data considers four Texas cities: Brownsville, Dallas, Austin, and El Paso. The aircraft type used by Hot Air is the Itty-Bitty-47, which holds a maximum of 120 passengers, and Hot Air has four of these planes. There is hangar capacity for two over-nighting planes in Brownsville and one each in Dallas and Austin. Three time periods per day are considered, each with its own demand for travel among the cities, see Table 6.14, Table 6.15 and Table 6.16. Ticket prices are given in Table 6.17. A ticket from Austin to Brownsville costs $109. The costs (per flight), which are the same across time periods, are given in Table 6.18. For instance, it costs $4400 to fly from Brownsville to Austin. This includes landing fees, crew costs, fuel etc.

Assignment 21.1

You have been hired to help Hot Air to optimize their daily schedule. Formulate a MIP model, implement it using Julia/JuMP, and present a solution to Hot Air management. During the flight to Hot Air's headquarters in the Windy City, you already realize that a possible set of decision variables would be $y_{t,i,j}$, which is the number of aircraft that fly during time period t from city i to city j, and $x_{t,i,j}$, the number of passengers being transported per time period t and from city i to city j. Of course, the latter is bounded by the number of people who want to fly, and the capacity of the planes flying on that route at that time. The data for the problem is given below.

Optimal for Hot Air 1: 12411

Assignment 21.2

Hot Air liked your solution. Now, however, they ask if there's a better way to assign planes to cities for overnight stays. In other words, could a higher profit be realized by changing the home ports of the planes? There's a one-time cost for any possible change (which will not matter in the long run), but operational costs will stay the same.

Hint: This is actually a simplification of the first model and can be solved very elegantly.

Optimal for Hot Air 2: 22368

Assignment 21.3

A manager at Hot Air wants to know about the actual profits for each flight and asks you to calculate these. Create a table that specifies the profit you make for each of the flights.

Assignment 21.4

The manager is shocked to find out that Hot Air is actually *losing* money on certain flights and decides that for now, no flights are allowed if they generate a negative profit. Reformulate your model from Assignment 21.2 to include this requirement. What happens?

Optimal for Hot Air 4: 20950

Assignment 21.5

Hot Air is beginning to feel the pressure from low-cost carriers and is looking for ways to compete with these. The marketing department discovers a new group of possible customers: Low-cost customers (e.g., students!) who can be attracted from bus transportation by lower prices and who do not expect premium service. The marketing department estimates that approximately an extra 50% of customers could be attracted if prices were lowered to 70% of the full-fair price. Hot Air does, however, *not* want to lower the price for the regular customers. Therefore, an employee from the marketing department comes up with the following idea: Only offer discount tickets at 70% of the price to customers who do *not* fly direct flights but have one stopover. Assume that you can still sell the full fair demand tickets, but also that you could additionally fill up the planes with discount passengers. Extend the model from the Assignment 21.2. Does this approach lead to higher profits?

This part of the Hot Air assignment is optional and rather hard to solve. If you are ambitious, go ahead !

Optimal for Hot Air 5: 28355.6

Hint: Remove the additional constraint from the previous question (engineers know better than managers) and consider introducing a variable to count passengers flying from city i to city j *via* city k. This refers to two flights, with the first flight in period t and the second in period $t + 1$. We will ignore discount travelers who wait for a period

or wait overnight. The new variable for the discount traveling passengers (probably students!) should then be added to the objective function *and* to the capacity constraints (they have to share planes with the normal passengers). When estimating the increased demand you should consider the time period of the first flight.

	Cities			
	Brownsville	Dallas	Austin	El Paso
Brownsville		50	53	14
Dallas	84		80	21
Austin	17	58		40
El Paso	31	79	34	

Table 6.14: Demand, period 1

	Cities			
	Brownsville	Dallas	Austin	El Paso
Brownsville		15	53	52
Dallas	17		134	29
Austin	24	128		99
El Paso	23	15	30	

Table 6.15: Demand, period 2

	Cities			
	Brownsville	Dallas	Austin	El Paso
Brownsville		3	16	9
Dallas	48		104	48
Austin	62	92		68
El Paso	13	15	21	

Table 6.16: Demand, period 3

	Cities			
	Brownsville	Dallas	Austin	El Paso
Brownsville		99	89	139
Dallas	109		99	169
Austin	109	104		129
El Paso	159	149	119	

Table 6.17: Ticket prices

	Cities			
	Brownsville	Dallas	Austin	El Paso
Brownsville		5100	4400	8000
Dallas	5100		11200	6900
Austin	4400	11200		5700
El Paso	8000	6900	5700	

Table 6.18: Cost per takeoff

6.22 Wind Farm Cable Routing

A new offshore wind farm that comprises 34 15MW turbines is being planned, and you have been asked to determine how best to connect the turbines to the substation, the location from which the generated power will be subsequently transformed and transferred to an onshore converter station. Turbines can be connected to another turbine or directly to the substation. A turbine can have any number of incoming connections, but is limited to at most one outgoing connection. The number of incoming connections (i.e., connected turbines) is limited by the capacity of the outgoing cable. There are three types of cables that can be used, and each differs in its capacity (i.e., the number of turbines it can support) and its cost per meter. The substation is limited to a maximum number of incoming connections. Your aim is to determine the cable types to use and the routing of each turbine to the substation such that the overall cost of the cable routing is minimized. Figure 6.3 gives an overview of the planned wind farm, complete with turbine locations (points 1-34) and the substation (point 35). You have been asked to consider two scenarios. In the first scenario, the substation can have at most 4 incoming connections, and the cable types are as specified in Table 6.19. In the second, the substation can have at most 3 incoming connections, and the cable types are as specified in Table 6.20. The distance matrix specifying the distance (in meters) between the locations of each pair of turbines is given in the file `wind_farm_data.txt`. This file also includes the coordinates of each turbine. In this version of the problem, we will assume that there is zero power loss in the cables.

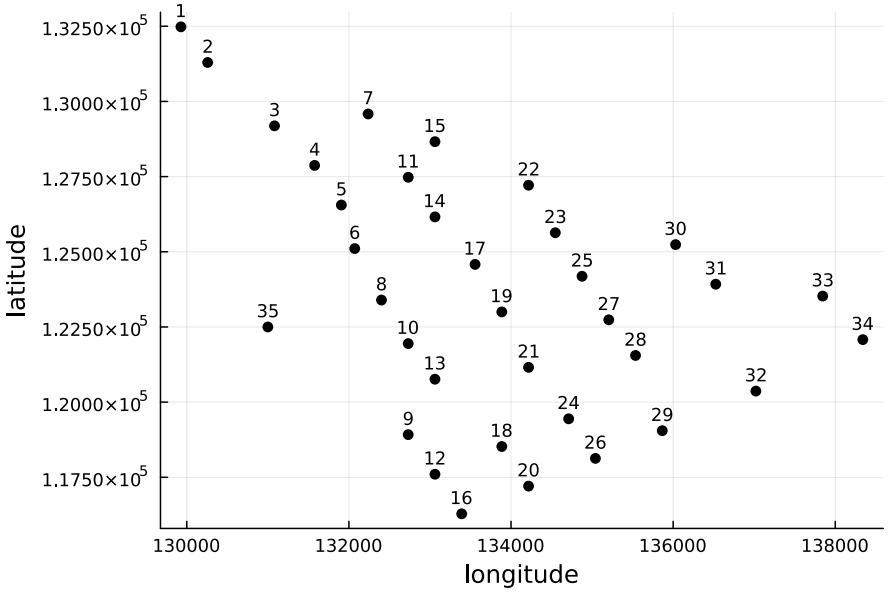


Figure 6.3: Wind farm overview

	Cable Type		
	1	2	3
Cost (€/m)	135	250	480
Max cable capacity (turbines)	2	6	9

Table 6.19: Cable Type Data - Scenario 1

	Cable Type		
	1	2	3
Cost (€/m)	170	290	610
Max cable capacity (turbines)	3	7	13

Table 6.20: Cable Type Data - Scenario 2

Assignment 22.1

Formulate a mathematical model that can be used to find an optimal cable routing for the planned wind farm. Implement your model using Julia/JuMP to find an optimal solution to each scenario.

Optimal objective value (Scenario 1): €12,058,455.40 (Gurobi: 18s, HiGHS: 203s)
Optimal objective value (Scenario 2): €12,791,103.40 (Gurobi: 25s, HiGHS: 260s)

Assignment 22.2

From an operational perspective, it is better if each "root branch" of the wind farm has more-or-less the same number of turbines, i.e., the routing is so-called *balanced*. A root branch can be thought of as the set of turbines routed to one connection point of the substation. Modify your formulation to ensure that the difference in the number of turbines between the root branch with the most turbines and the root branch with the least turbines is at most one. Implement your model using Julia/JuMP to find an optimal solution to each scenario.

Optimal objective value (Scenario 1): €12,147,938.10 (Gurobi: 8s, HiGHS: 188s)
Optimal objective value (Scenario 2): €12,795,645.80 (Gurobi: 17s, HiGHS: 114s)

	Cable Type		
	1	2	3
Cost (€/m)	135	290	610
Max cable capacity (turbines)	4	6	12

Table 6.21: Cable Type Data - Scenario 3

Assignment 22.3

It is highly undesirable that cables between turbines cross each other. Using the data given in Table 6.21 and assuming that the substation can have at most three incoming connections, verify that in an optimal solution the cables between certain turbines do in fact cross. Identify, a constraint that can be added to the model to ensure that cables do not cross each other. Implement your model using Julia/JuMP to find an optimal solution in which cables do not cross. Assume that root branches need not be balanced.

Optimal objective value (crossings): €10,574,236.65 (Gurobi: 6s, HiGHS: 67s)
Optimal objective value (no crossings): €10,581,449.70 (Gurobi: 4s, HiGHS: 37s)

6.23 University Timetabling

Creating timetables for universities is a complicated planning task. How the timetables are planned differs a lot from university to university, but in this assignment we will use data from the second university timetabling competition from 2007. These data comes from the University of Udine, Italy. During the questions in this assignment, the model will be expanded, ending up with the full model. Notice: The models becomes rather hard and the HiGHS solver will struggle. The final model can only be solved by the Gurobi solver **and** only for some of the problems.
The basic assignment is to assign lectures to rooms and time slots, such that all lectures are planned. The basic data for the assignment are:

Sets of entities:

- C: No of courses
- L: No of lecturers
- R: No of rooms
- D: No of days
- P: No of periods each day
- CU: No of curricula

Data:

- $penalty\_missing\_seat$: Penalty for each seat missing for the students of course c
- $penalty\_room\_change$: Penalty for each room more than 1 for each course c
- $penalty\_less\_than\_minimum\_working\_days$: Penalty for each teaching day less than $Course\_MinLectureDays_c$ for course c
- $penalty\_solo\_cur\_courses$: Penalty for each isolated course c for each curricula
- $Course\_NoStudents_c$: No of students for course c
- $Course\_NoLectures_c$: No of lectures for each course c
- $Course\_Lecturer_c$: Index of lecturer teaching course c
- $Lecturer\_NoCourses_l$: No of courses for lecturer l
- $Rooms\_size_r$: Size of room r

- $Course\_MinLectureDays_c$: Minimum number of days required for teaching of course c
- $Curricula\_CourseCurreculaIncidenceMatrix_{cu,c}$: Incidence matrix, which is 1, if course c is part of curricula cu

From DTU-learn, you can download 3 out of the 21 original dataset, the dataset $timetable1.jl$, $timetable4.jl$ and $timetable5.jl$. These datasets can be directly included into the Julia program and contains the above data.

The basic feasibility requirement is that all lectures of all courses can be carried out. Notice, that it is accepted that there are not enough seats to the students. The requirements are:

- All lectures of all courses should be allocated
- At most one lecture can take place in a room for each timeslot on each day
- A lecturer can at most teach one course in each timeslot on each day
- Only one course for a curricula may be taught in each timeslot on each day
- For each course, there are a number of blocked timeslots, i.e. days and periods where the course cannot be taught by the lecturer. If $Blocked\_course\_timeslot[c, d, p] = 1$, then course c cannot be taught on day d in period p. You can either formulate this as constraints, or use the fix command.

Assignment 23.1

Formulate the feasibility model. If necessary declare the objective constant.

Optimal objective value simply equal to what constant you have put in the objective function

Generally, our students like to sit down during the lectures! To avoid the risk of infeasibility we will allow to teach courses with more students than seats in the , but penalize each missing seat for each lecture in each course. The size of the penalty $penalty\_missing\_seat$ is given in the data.

Data set	Is optimal solution found	Time used in seconds	Best found solution	Best lower bound
timetable1.jl	Optimal	0.7	42.0	42.0
timetable4.jl	Optimal	3.6	42.0	42.0
timetable5.jl	Optimal	1.9	42.0	42.0

Table 6.22: Table of result for Assignment 22.1. Solution time, using HiGHS on a Mac Book Pro (M2). Notice that when the found solution is optimal, then the best found solution objective and the lower bound is identical

Assignment 23.2

Minimize the number of missing seats for each course.

Data set	Is optimal solution found	Time used in seconds	Best found solution	Best lower bound
timetable1.jl	Optimal	3.4	4.0	4.0
timetable4.jl	Optimal	6.8	0.0	0.0
timetable5.jl	Optimal	5.4	0.0	0.0

Table 6.23: Table of results for Assignment 22.2. Solution time, using HiGHS on a Mac Book Pro (M2). Notice that when the found solution is optimal, then the best found solution objective and the lower bound is identical

For both students and lecturers it is annoying if different lectures in a course is taught in different rooms. Hence the second objective is to minimize the number of used rooms for the courses. Naturally, each course will need to use at least one room. If **more** than one room is used, then the number of extra rooms are pelalized, with the penalty $penalty\_room\_change$.

Assignment 23.3

Minimize the number of missing seats for each course **and** the number of extra rooms for each course, added with the correct penalty terms

For some courses, the lecturers request that the lectures are spread over the week, e.g. if a course have 4 lectures, they should at least be taught over 3 days. If all the lectures of

Data set	Is optimal solution found	Time used in seconds	Best found solution	Best lower bound
timetable1.jl	Optimal	35.1	5.0	5.0
timetable4.jl	Optimal	176.5	0.0	0.0
timetable5.jl	Optimal	40.9	0.0	0.0

Table 6.24: Table of results for Assignment 22.3. Solution time, using HiGHS on a Mac Book Pro (M2). Notice that when the found solution is optimal, then the best found solution objective and the lower bound is identical

the course is given the same day, the difference between the wished number of teaching days of the course and the actual number of teaching days, is 2. This is then multiplied with the penalty parameter *penalty_less_than_minimum_working_days*

Assignment 23.4

Minimize the number of missing seats for each course, the number of extra rooms for each course **and** the penalty for the less than the number of required teaching days.

Data set	Is optimal solution found	Time used in seconds	Best found solution	Best lower bound
timetable1.jl	Optimal	119.0	5.0	5.0
timetable4.jl	Optimal	557.6	0.0	0.0
timetable5.jl	Optimal	41.7	15.0	15.0

Table 6.25: Table of results for Assignment 22.4. Solution time, using HiGHS on a Mac Book Pro (M2). Notice that when the found solution is optimal, then the best found solution objective and the lower bound is identical

Finally, students do not like to have spare time between teaching (do you?). This can be modelled in different ways, here we do the following: Here we want to avoid **isolated** curricula courses, i.e. courses for which no other courses for the same curricula are taught either before or after. For each isolated course for each curricula there is a penalty of *penalty_solo_cur_courses*. **NOTICE:** With this extension the model becomes hard to solve, even for Gurobi.

Assignment 23.5

Minimize the number of missing seats for each course , the number of extra rooms for each course, the penalty for the less than the number of teaching days **and** the penalty for isolated curricula courses.

The addition of the last criteria in the objective function, makes the problem much harder. As can be seen in the two tables below, HiGHS only finds **a** solution for the smallest dataset, timetable1.jl, and this solution is not optimal. For timetabl4.jl and timetable5.jl, -1 is reported, corresonding to not even finding a feasible solution. Gurobi, on the other hand, finds optimal solutions to both timetable1.jl and timetable4.jl, but fails to find the optimal solution to timetable5.jl. Gurobi does however find a feasible solution, but this solution has a large gap between the best possible (bound) objective value (149) and the found solution (371).

Data set	Is optimal solution found	Time used in seconds	Best found solution	Best lower bound
timetable1.jl	Not optimal	600.0	17.0	4.0
timetable4.jl	Not optimal	600.1	-1.0	7.1
timetable5.jl	Not optimal	600.1	-1.0	67.4

Table 6.26: Table of results for Assignment 22.5. Solution time, using HiGHS on a Mac Book Pro (M2). Notice that when the found solution is optimal, then the best found solution objective and the lower bound is identical

Data set	Is optimal solution found	Time used in seconds	Best found solution	Best lower bound
timetable1.jl	Optimal	12.5	5.0	5.0
timetable4.jl	Optimal	486.3	35.0	35.0
timetable5.jl	Not optimal	600.0	371.0	149.0

Table 6.27: Table of results for Assignment 22.5. Solution time, using HiGHS on a Mac Book Pro (M2). Notice that when the found solution is optimal, then the best found solution objective and the lower bound is identical

7 Multi-Objective Integer Programming

The Incredible Chairs company has a new challenge: Because of the general focus on climate all furniture is now rated according to it's CO2 emissions, on a scale from 1 to 10, where 10 is the best (lowest CO2 emissions) and 1 is the worst (highest CO2 emissions). How can the production be planned, taking into account **both** profit and climate rating ?

Recall the Incredible Chairs 3 MIP model, given $c \in C$ chairs where production is limited by the production capacity of $p \in P$ production lines. For each whole chair produced there is a profit of $Profit_c$. Each production line has a capacity limitation of $Capacity_p$, where each chair of type c produced requires $RecourseUsage_{c,p}$ capacity of production line p. Our variables $x_c \in Z^+$ specify how many of each chair should be produced. The MIP model for maximizing the Profit is given below:

$$\begin{array}{ll} \texttt{Maximize.} & \sum_c Profit_c \cdot x_c \\ \texttt{Subject to :} & \\ & \sum_c RecourseUsage_{c,p} \cdot x_c \ \leq \ Capacity_p \ \forall p \\ & x_c \in Z^+ \end{array}$$

The new CO2 rating is given below in Table 7.1.

	Chair types									
	A	B	C	D	E	F	G	H	I	J
Rating	4	5	1	5	4	7	6	3	6	7

Table 7.1: CO2 Rating for each chair

To find a good production plan where both profit and rating is taken into account, a new model is formulated with **two** objective functions. See the model below.

$$\begin{array}{ll} \texttt{Maximize.} & \sum_c Profit_c \cdot x_c \\ & \sum_c Rating_c \cdot x_c \\ \texttt{Subject to :} & \\ & \sum_c RecourseUsage_{c,p} \cdot x_c \ \leq \ Capacity_p \ \forall p \\ & x_c \in Z^+ \end{array}$$

In the new model, there is no longer **one** unique best solution, but a whole **set** of solutions, in the Incredible chairs case the following solutions are all best, see the list below in Table 7.2.

Solution no	Profit	Rating
Solution 1	18	12
Solution 2	17	13
Solution 3	16	14
Solution 4	15	25
Solution 5	13	27

Table 7.2: The 5 optimal solutions regarding profit and ranking

In Table 7.2 there are 5 solutions, which each are **Pareto optimal**. By this we informally mean that for each of these 5 solutions no other solution exists which are better (higher) in one of the objectives and at least as good on the other. Which of these 5 different production plans to choose is then up to the Decision Maker (DM), who must prioritize what is best for the company, pure profit, then solution 1 is best or if CO2 emissions are most important, then Solution 5. Hence, the optimal solution to a Multi-Objective Mixed Integer Programming model is a **set** of solutions, here 5.

Thanks to a very recent development, February 2023, it is now possible to formulate multi-objective mathematical programming models in Julia/JuMP. The model used to find the set of 5 solutions above for the multi-objective incredible chairs problem is given below.

Multi-Objective Incredible Chairs Julia/JuMP program:

```
#************************************************************************
# Multi-Objective Incredible Chairs 4
using JuMP
using HiGHS
using MultiObjectiveAlgorithms
#************************************************************************

#************************************************************************
# Data
Chairs=["A", "B", "C", "D", "E", "F", "G", "H", "I", "J"]
C=length(Chairs)
ProductionLines=[1, 2, 3, 4, 5]
P=length(ProductionLines)
```

```
14
15 Profit=  [6, 5, 9, 5, 6, 3, 4, 7, 4, 3]
16 Rating = [4, 5, 1, 5, 4, 7, 6, 3, 6, 7]
17
18 Capacity=[47, 19, 36, 13, 46]
19 RecourseUsage=[
20 6  4 2  3 1 10 2 9  3 5;
21 5  6 1  1 7  2 9 1  8 6;
22 8 10 7  2 9  6 9 6  5 6;
23 8  4 8 10 5  4 1 5  3 5;
24 1  4 7  2 4  1 2 3 10 1]
25 ##########################################################################
26
27 ##########################################################################
28 # Model
29 MO_IC = Model()
30 @variable(MO_IC, x[1:C]>=0, Int)
31
32 # Objective 1: Maximize Profit
33 @expression(MO_IC, Profit_expr, sum(Profit[c] * x[c] for c=1:C))
34
35 # Objective 2: Maximize Rating
36 @expression(MO_IC, Rating_expr, sum(Rating[c] * x[c] for c=1:C))
37
38 # Define combined BI-OBJECTIVE
39 @objective(MO_IC, Max, [Profit_expr, Rating_expr])
40
41 # Capacity limitations
42 @constraint(MO_IC, [p=1:P], sum( RecourseUsage[p,c] * x[c] for c=1:C) <=
   ↪  Capacity[p])
43 ##########################################################################
44
45 ##########################################################################
46 # Solve
47
48 # Set MIP solver
49 set_optimizer(MO_IC, () -> MultiObjectiveAlgorithms.Optimizer(HiGHS.Optimizer))
50 set_silent(MO_IC)
51
52 # Set MO solver
53 set_attribute(MO_IC, MultiObjectiveAlgorithms.Algorithm(),
   ↪  MultiObjectiveAlgorithms.EpsilonConstraint())
54
55 optimize!(MO_IC)
```

```
56 #solution_summary(MO_IC)
57 ##########################################################################
58
59 ##########################################################################
60 # Solution
61 if result_count(MO_IC)>=1
62     println("No Pareto optimal points: ",result_count(MO_IC))
63     for i in 1:result_count(MO_IC)
64         y = objective_value(MO_IC; result = i)
65         println("Objective 1: ",round.(Int,y[1]), "    Objective 2:
           ↪  ",round.(Int,y[2]))
66     end
67
68 else
69     println("No solutions found")
70 end
71 ##########################################################################
```

We will now make a number of comments to this above Julia/JuMP program:

- The first part, line 1-6, imports the standard packages JuMP and HiGHS, but also a new one: MultiObjectiveAlgorithms. This package is necessary for the multi-objective part of the problem.
- The second data part, line 8-25, is similar to the previous Incredible Chair 3, only adding only one extra data: Rating.
- The model part, line 27-43, is slightly different:
  - Declaring the variables is as usual, line 30
  - Declaring the constraints is as usual, line 42.
  - Declaring the **objective function** is different. First an expression for each objective in line 33 and line 36. Finally, the two expressions are declared to be a vector objective function, line 42.
- The solve part, line 45-57, is significantly different. First the MultiObjectiveAlgorithms optimzer is told to use the HiGHS solver as a sub-routine in solving the multi-objective model. Then one of the available bi-objective optimization algorithms offered in the MultiObjectiveAlgorithms is selected for this problem in line 53. With these definitions, the problem can be optimized.
- The solution part, line 59-71, is slightly different, since what is returned is not just one solution, but a whole list of solutions. The number is found using the result_count function.

7.1 The importance of Multi-Objective models

When solving real-world problems it quickly becomes clear that the real-world is multi-objective. When humans look at solutions to planning problems, they will most often evaluate the solutions according to a list of different measures (objectives).

- Project scheduling. Here there are two objectives: Minimize the used money and minimize the used time.
- Chair Logistics 2. Here there are two costs which are minimized, transport costs and depot costs. Even though both costs are measured in money, some of the costs are bound as fixed costs in the depots whereas other costs, the truck transportation costs, are variable.
- Tariff Rates. At first glance, only one objective is used, money. But the requirement that at least 15 % extra demand should be possible to be produced by the running plants. These 15 % could instead be considered a second objective, to be maximized.
- The Wedding Planner. Here there are 3 objectives, maximize the shared topics, minimize the gender imbalance at the different tables and minimize the number of people with no friends at the tables.
- Hot Air. Here the objective is money, but there are two different types of tickets: Regular and discount tickets. Both objectives can be maximized.

7.2 Multi-objective theory

Multi-objective optimization has been researched for at least the last 50 years. The state of both the theory and the solvers available for practitioners is unfortunately less mature as single objective optimization. Given the importance of multi-objectiveness in real-life problems, we decided to include this in the course. We would however like to stress that this chapter provides only a very brief introduction. We would also like to stress that the optimization algorithms available in Julia/JuMP may still be somewhat unreliable.

In this course, we utilize a very recent addition to the JuMP package: In March 2023 the possibility to model with vector objective functions was introduced. To the best of our knowledge, this is the first modelling language where this is possible.
The general multi-objective Mathematical Program we consider in this course is given below. In this course, we only consider multi-objective Mathematical Program with linear objective functions and linear constraints (as illustrated below).

$$
\begin{array}{ll}
\texttt{Maximize/Minimize} & \sum_i F_i^1 \cdot x_i \\
& \sum_i F_i^2 \cdot x_i \\
& \ldots \\
& \sum_i F_i^k \cdot x_i \\
\texttt{Subject to :} & \\
& \sum_i A_{i,j} \cdot x_i \quad \leq \quad B_j \ \forall j \\
& x_i \in R, Z, \{0,1\}
\end{array}
$$

The model above, has k objective functions. We will assume that all the objective functions are either maximized or minimized. Notice, that to achieve minimization of an objective function is the same as maximizing the function multiplied by -1. A very important point is that there (usually) is usually not just one optimal solution but many.

To compare solutions we introduce the concept of *(Pareto) dominance*: A feasible solution $\mathbf{x}$ *(Pareto) dominates* another solution $\mathbf{y}$ if $f_i(\mathbf{x}) \leq f_i(\mathbf{y})\ \forall i \in \{1, 2, \ldots, k\}$ and $f_j(\mathbf{x}) < f_j(\mathbf{y})$ for at least one $j \in \{1, 2, \ldots, k\}$ (assuming minimization). Thus, no other solution can dominate a Pareto optimal solution. We will use the terms *Pareto dominant solutions* or *non-dominated solutions* interchangeably. When solving a multi-objective optimization program like the one above, the goal is to find the set of all non-dominated solutions. This set we call the Pareto Front.

There are many different types of multi-objective mathematical programming. They depend on a number of parameters:

- The number of objectives
- Whether all variables are continuous. Then the model will be a Multi-Objective Linear Program (MOLP)
- Whether there are continuous variables in two or more objective functions
- If the objective functions only contain integer variables **and** all the objective function coefficients are integer ($F_i^1 \in Z$)

In this book we only consider the simplest versions of the multi-objective mathematical programming models. Hence we will restrict our attention to multi-objective optimization with two objective functions. Furthermore, we will only consider problems where the pareto front only consists of points, which is illustrated below.

For multi-objective problems with only two objectives we can illustrate different types of pareto fronts. If the multi-objective mathematical programming model only contains continuous variables, i.e. is a MOLP model, the Pareto Front becomes continuous, see

Figure 7.1, where we assume maximization. In this case there is an **infinite number** of solutions on the Pareto Front, corresponding to the connected line Figure 7.1.

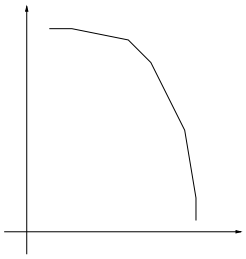


Figure 7.1: A continuous Pareto Front

If a multi-objective **mixed integer** programming model, where one of the objective functions only contains integer/binary variables, then the Pareto Front becomes point-wise, see Figure 7.2, where we assume maximization. In Figure 7.2 the Pareto Front consists of a finite number of Pareto points, corresponding to the dots.

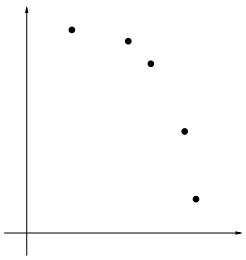


Figure 7.2: A point-wise Pareto Front

When performing multi-objective optimization it is much easier to consider point-wise Pareto Fronts, and we will in this course **only** consider point-wise Pareto front problems.

A final simplification is sometime present in the models. Beside being point-wise, if only integer/binary variables are in the objective functiosn **and** all the coefficients in the objectives are integer, i.e. $F_i^d \in Z$, then the objective functions can naturally only take integer values. In Figure 7.3 it is illustrated that all the point solutions are placed in a grid. Having only integer valued Pareto solutions often simplifies the work for the multi-objective optimization algorithms.

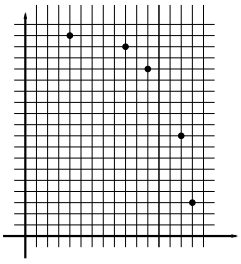


Figure 7.3: A point-wise Pareto Front in an integer grid

7.3 Multi-objective optimization algorithms

As mentioned earlier, access to multi-objective optimization through JuMP is a very new thing and the software is still under construction. Using JuMP, trough the Julia package MultiObjectiveAlgorithms, 6 different algorithms are offered to the users.

The main one and the only one we will use in this course is the ϵ-Constraint algorithm. In the above code it is setup in line 49 and line 53. The ϵ-Constraint algorithm use a MIP solver as a sub-routine, and which solver is used (described in more detail in the section below), is setup in line 49. This setup line is necessary for all the multi-objective algorithms available through JuMP. The second part is that actual choice of which multi-objective algorithm to use, is given in line 53. The alternatives to the ϵ-Constraint algorithm are:

- Epsilon Constraint Chankong and Haimes (2008). This is the classic ϵ-constraint method, invented in 1983.
- Dichotomy Aneja and Nair (1979). Notice that this method does **not** find the full pareto front, only the extreme points of the pareto front, i.e. the solutions which can be found using different secularization's of the objectives.
- Chalmet Chalmet et al. (1986)
- DominguezRios Domínguez-Ríos et al. (2021)
- KirlikSayin Kirlik and Sayın (2014)
- TambyVanderpooten Tamby and Vanderpooten (2021)

7.4 The epsilon constraint algorithm

The simplest multi-objective optimization algorithm to solve a two dimensional multi-objective problem is the **epsilon-constraint** (ϵ-constraint) algorithm. The basic layout is given below.

Algorithm 1: The ϵ-Constraint algorithm

```
1 solution-exists=true;
2 while solution-exists do
3     (solution-exists,z1,z2)=LexioGraphicZ2Z1();
4     if solution-exists then
5         AddParetoPoint( (z1,z2) );
6         SetLowerBoundZ1(z1+δ);
```

In the pseudo-code above an ϵ-Constraint algorithm for a bi-objective maximization problem is shown. First a boolean variable is set to true (line 1). This variable is used to determine if the algorithm should terminate. Then the main while loop is executed (line 2). Then a function, which performs lexiographic optimization is called: First maximize regarding z2, then add a lower bound on z2, such that it cannot become worse, smaller than the optimal current value, and optimize (maximize) regarding z1. If no lexiographic solution is found the Lexiographic function should return the value **false** to the solution-exist variable, and the algorithm will terminate. If a solution with the current limitation exists, it is part of the front and is saved (line 5). Finally, the lower bound on z1 is increased, above the z1 value of the current solution (line 6). If the problem is a point wise problem in an integer grid, see Figure7.3, then setting $\delta > 0 \wedge \delta < 1$ will lead to an optimal Pareto front, given enough time.

We illustrate the ϵ-Constraint algorithm graphically below in 4 steps in Figure 7.4 to Figure 7.7. In Figure 7.4 the leftmost Pareto point is found, called the Upper Left (UL) Pareto point. In Figure 7.5, a lower bound on z1 is added, making UL infeasible. In Figure 7.6 the lexiographic optimization, z2 first then z1 is used to find the next point. This continues until the Lower Left (LL) Pareto point is found. When it is found, we however do not know that it is found. Only, when another bounding constraint is added, making LL infeasible and lexiographic optimization, z2 first then z1 is used, and it returns infeasible, we know that the LL has been found, and the Pareto front is complete.

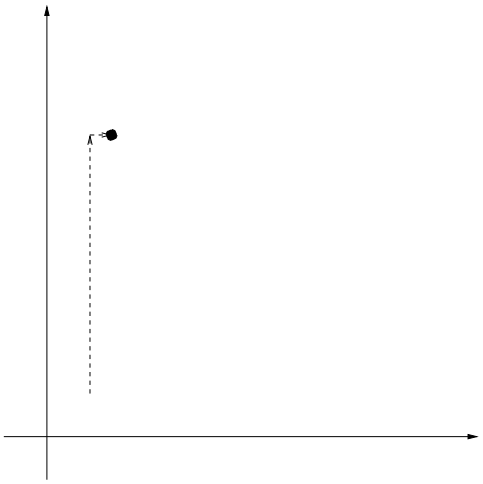


Figure 7.4: Finding Upper-Left (UL) Pareto point through lexiographic z2 $\rightarrow$ z1 optimization

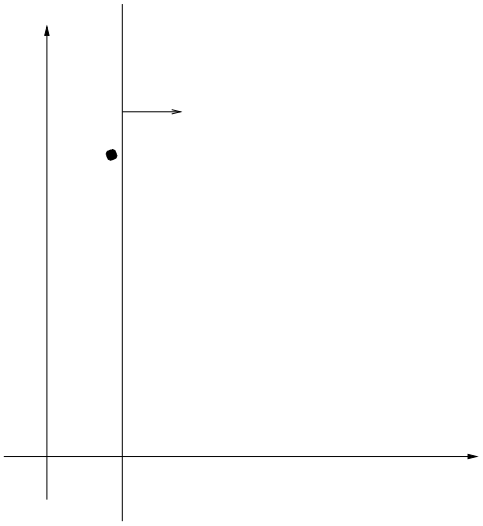


Figure 7.5: Adding bounding constraint on z1

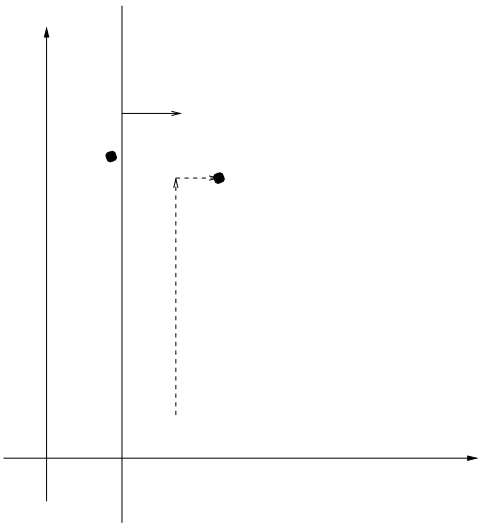


Figure 7.6: Finding next Pareto point through lexiographic z2 → z1 optimization

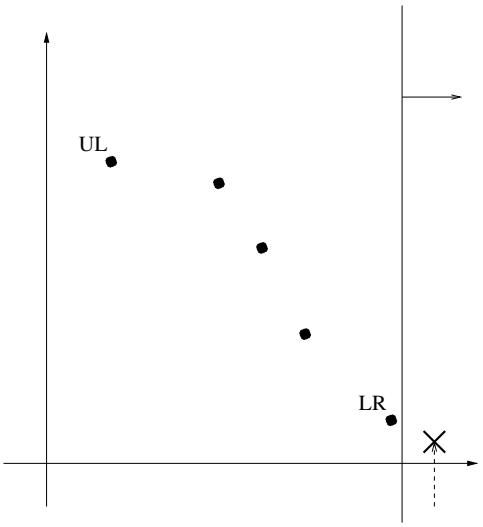


Figure 7.7: Finding **no** solution after last point, Lower Left (LL).

A few implementation advices when implementing the ϵ-Constraint algorithm in Julia/JuMP:

- In the Lexiographic function, it is necessary to change the objective function, to first maximize z2 and then maximize z1. This is actually easy: Whenever a new "@objective" command is issued, the current objective function is replaced with the new version.
- Lower limiting the value of the z1 variable is also relatively simple: Either set the lower bound on the variable, or add a constraint. Notice that the previous

constraints are not removed, but have a lower right-hand-side value for z1, and can hence be ignored.

8 Multi-Objective optimization Programming problems

8.1 Multi-objective Startup Fund

The manager of the Startup Fund manager now hear about the 17 Sustainable Development Goals (SDG) and start to wonder how SDG friendly his investment is. Hence he hires an Non-Governmental Organsation (NGO) to evaluate the 9 investments according to the Sustainable Development Goals (SDG) and rate the different investment on a scale between 1 (worst) and 10 (best).

	Investment								
	1	2	3	4	5	6	7	8	9
SDG	8	6	8	3	4	5	3	2	7

Table 8.1: SDG rating from 1 to 10

You might wonder why the investment manager bothers! Now he may be personally invested in SDG, but really it is about **risk** avoidance. If he invest in a company which ruins the clean water in it's neighbourhood (not fulfilling SDG 6) there is a high probability that at some point there is going to be blow back either from government or from the local population. And this is probably bad for business ! There are 17 different goals (`https://sdgs.un.org/goals`) so it is very difficult to evaluate all of them, hence the overall rating according to all the goals. In this version of Startup fund, we use the **first** version without the extra demands to the composition of the investment.

Assignment 1.1

Find the best possible SDG value.

Optimal objective value for Startup Fund 2, SDG: 29

Assignment 1.2

Now the startup investment manager becomes worried, maybe the good SDG solution can be improved regarding the profit. Hence he wants the Max SDG and then Max profit solution, called the Lexicographic SDG-profit solution.

The Lexiographic SDG-profit solution, SDG: 29 and profit: 116. Notice, that this is actually the same result as in the previous assignment. But there we could not know that there would actually not be a better profit, with an SDG of 29. Now we know it to be this way.

Assignment 1.3

Then look at the other extreme: The Lexicographic profit-SDG solution.

The Lexiographic SDG-profit solution, profit: 191 and SDG: 19
To get the best data for making the investment decision, the startup fund manager now requests the **full** Pareto front, where the Lexicographic points are the two extreme points.

Assignment 1.4

Find the entire Pareto front, regarding SDG objective and profit objective

Optimal bi-objective Pareto Front

		Investment								
		1	2	3	4	5	6	7	8	9
Profit Obj	SDG Obj	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
116	29	1	1	1	0	0	0	0	0	1
122	28	1	0	1	0	0	1	0	0	1
134	27	1	0	1	0	1	0	0	0	1
141	26	1	1	1	0	1	0	0	0	0
149	25	1	0	1	0	0	0	0	1	1
156	24	1	1	1	0	0	0	0	1	0
162	23	1	0	1	0	0	1	0	1	0
174	22	1	0	1	0	1	0	0	1	0
178	21	1	0	0	0	1	0	0	1	1
191	19	1	0	0	0	1	1	0	1	0

Table 8.2: Profit-SDG Pareto front

8.2 Multi-objective Young Couple 2

Eve and Steven realize that their original work distribution problem is actually multi-objective (bi-objective): The want to minimize the total amount of time used on the work **and** they want to ensure that the time each of them spend is as similar as possible. Furthermore, they want to include additional work, on top of the 4 jobs given in the first version of the assignment. The work time for the new jobs is given in Table 8.3:

	lawn	shopping	carwash
Eve	2.1	1.9	2.5
Steven	1.5	3.2	1.2

Table 8.3: Task times (hours)

Assignment 2.1

Convert the model to a bi-objective model, where both the total worktime and the difference in worktime (always positive), is minimized.

Total work time	Work time difference	Eve worktime	Steven worktime
22.9	0.7	11.1	11.8
24.0	0.2	12.1	11.9
25.2	0.2	12.5	12.7
27.0	0.0	13.5	13.5

Table 8.4: List of 4 different Pareto optimal work-plan solutions

Optimal objective value for Young Couple Problem: 18.2 hours

With the list of Pareto optimal solutions in Table 8.4 Eve and Steven can now ponder on a classical planning problem: **What is the value of equality?**

8.3 The Multi-objective Wedding Planner

The wedding planner had 3 objectives:

1. Maximize the sum of shared interests at each table
2. Minimize the difference in the number of men and women at each table above 2
3. Minimize the number of guests who do not know at least 3 at their table

In this assignment you should solve the problem bi-objectively: Maximize the sum of the shared interests and maximize the sum of the **negative** number gender differences plus the **negative** number of guest who know less than 3.

Assignment 3.1

Find the entire Pareto front, for the 2 objective version

Optimal bi-objective Pareto Front

Shared Interests	Negative Gender + Know
Objective 1: 48	Objective 2: -2
Objective 1: 60	Objective 2: -5
Objective 1: 63	Objective 2: -5
Objective 1: 64	Objective 2: -5
Objective 1: 65	Objective 2: -5
Objective 1: 66	Objective 2: -8
Objective 1: 67	Objective 2: -8

Table 8.5: Pareto front, found using MultiObjectiveAlgorithms.EpsilonConstraint(). Notice the algorithm returns 4 dominated solutions

8.4 Multi-objective Aircraft Landing

Let us now consider a bi-objective variant of Assignment 6.16. While it is extremely important to ensure that the aircraft arrival times deviate as little as possible from their respective target arrival times, it is important to ensure that the time between consecutive aircraft landings is as large as possible (to prevent the propagation of any unforeseen delays). Changing the landing sequence is seen as undesirable as ground operations at the airport are made aware of, and anticipate, a certain landing sequence.

Assignment 4.1

Modify your formulation from Assignment 16.2 (assume that aircraft 5 and 6 can still not land consecutively in any order), so that it now maximizes the minimum separation time between consecutive aircraft landings. Implement your model in Julia/JuMP to find an optimal solution to this *single-objective problem.*

Optimal objective value: 53

Assignment 4.2

You observe that with this formulation, the penalty associated with target time deviation can be arbitrarily large. By modifying your formulation, find the landing sequence with the total minimum penalty, subject to the requirement that the separation must be at least 53 minutes between any two consecutive landings.

Optimal objective value: 46525

Assignment 4.3

Now consider the bi-objective version of the problem in which you simultaneously try to minimize the total penalty incurred and maximize the minimum separation time between any two consecutive aircraft landings. Explore the trade-off between the two objective functions by implementing your model in Julia/JuMP to obtain the entire pareto. Try plotting the pareto front. **Note: Gurobi is the suggested solver for this assignment. HiGHS will work but be prepared to wait some time!**

Number of pareto optimal solutions: 49

8.5 ϵ-Constraint algorithm

Implement the ϵ-Constraint algorithm described in Section 7.4, and apply it to one of the previously implemented Multi-Objective problems, e.g. Startup Fund, Young Couple, Wedding Planning or Aircraft Landing.

9 Data

In Julia there are many different data types. The data types we need in this book for our LP models and MIP models are simple: Scalar numbers and arrays of numbers of different size and dimensionality.

9.1 Direct data definition

In the assignments, we need arrays of different dimensions:

- Scalar numbers (zero-dimensional arrays):

```
f=90
Q=31
```

- Vectors (one-dimensional):

```
customer_orders=[11 7 6 12]
C=length(customer_orders)
```

  Notice: The space character separates the numbers. Here we have a vector with 4 numbers. This number can be found using the length function. To access the 3 element simply write:

```
customer_orders[3]
```

  The index always starts at 1 and ends at the length of the array, here 4. If an attempt is done to access an element outside the array, e.g. 0, Julia will give an error.

- Matrix's (two-dimensional):

```
distances=[
17  39  84 ;
109  32  98 ;
```

```
139   4  31 ;
123 129 116 ]
(I,J)=size(distances)
```

Notice: Again the space character separates the numbers, but additionally the semicolon separates the different rows. The size function can be used to find that there are 4 rows (I=4) and 3 columns (J=3). To access the 3 element of the 4 row write:

```
distance[4,3]
```

which has the value 116. For each dimension index validity is checked.

- Matrix's (three-dimensional):

```
time_distances=zeros(Int16,2,4,3)
time_distances[1,:,:]=[
17  39  84 ;
109  32  98 ;
139   4  31 ;
123 129 116 ]
time_distances[2,:,:]=[
37  49  54 ;
609  72  88 ;
939   1  21 ;
133 149 156 ]
(T,I,J)=size(time_distances)
```

Notice: This time, firstly a 3-dimensional matrix of zero values is created. Then the two two-dimensional sub-matrices are filled out. Again we can access elements:

```
time_distance[2,2,3]
```

9.2 Including Julia data

The data section in a Julia program can become rather large and it may be convenient to store the data in a separate file. This can be done very simply by:

```
include("NetworkData.jl")
```

Notice that the data file has to have a Julia format and it is simply inserted into the Julia file at the place of the include statement. Notice that you may have to write the full path to the file.

10 Future reading

This book aims at teaching basic Mathematical Programming modelling, both for LP models and MIP models. However, there are a lot of important subjects regarding Mathematical Programming and the broader field of Operations Research that have been omitted. In this section, we will try to give pointers to material for further studies.

The most used introductory textbook globally on Operations Research is "Introduction to Operations Research" Hillier and Lieberman (2020). This book introduces a lot of topics in Operations Research, including LP and MIP. This book offers a good starting place for anyone who would like a broader introduction to Operations Research.

For a good and more mathematical introduction to the mathematics behind solving MIP, we recommend "Integer Programming" Wolsey (2020).

In our opinion, a good book introducing Julia as a programming language, is still missing. The best material is available online at `julialang.org`, which is also kept up to date.

10.1 Classic OR problems

Most real-world optimization problems are both unique and often very similar. The similarity arises by including parts of a list of classical, well-known Operations Research problems. A number of the exercises in this book are actually formulations of these classic problems, see the list below:

- **The Project Scheduling Problem**: Project Scheduling 4.6
- **The Knapsack Problem**: Startup Fund 6.2.
- **The Set Packing/Set Partitioning Problem**: Stamps 6.4
- **The Bin Packing Problem**: Scrap Removal 6.10
- **The Graph Coloring Problem**: Food Festival 6.17

- **The Lot Sizing Problem**: Micro Brewery 2 6.5
- **The Chinese Postman Problem**: Tennis 6.14
- **The Travelling Salesman Problem**: Groceries 6.15
- **The Vehicle Routing Problem**: Groceries 6.15

10.2 Optimization hardness

As mentioned in Chapter 7, the required solution times for MIP models above a certain problem size increases dramatically. The theoretical explanation for this increased solution time is based on the computational complexity theory $P \neq NP$ Garey and Johnson (1979). While computational complexity theory deals with **decision problems** and MIP models handles optimization problems, many of the MIP models are generalizations of so-called NP-complete decision problems. Therefore, the exponential increase in solution time of MIP models is unavoidable. The good news is, however, that tremendous improvements have been a achieved in the last decades Bixby (2012), so that even though a problem may be theoretically hard to solve, MIP models of the relevant size can be solved. Unfortunately, it is impossible to predict how hard it is for a MIP model solver to solve a problem because the complexity of the solvers has grown significant and so has their behaviour on different MIP models, despite the improved average performance Lodi (2013). The easiest way to find out whether a MIP model can be solved is simply to implement it.

Author biographies

Richard Lusby is an Associate Professor at the Technical University of Denmark and teaches courses in Operations Research, Mathematical Modelling, and Decomposition Methods. His main research interests include Integer Programming, Decomposition Methods, and Matheuristics, and their application to public transportation and manpower planning problems.

Thomas Stidsen is an Associate Professor at the Technical University of Denmark and teaches courses in Mathematical Modelling, Metaheuristics, and Decomposition Methods. His main research interests include optimization approaches for educational timetabling and manpower planning, and multi-objective optimization.

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